JESS Thermodynamic Database v8.9

Zirconium

24-Jan-23

Reaction No. 2739							
Zr+4 + H+1_F-1 = Zr+4_F-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.0(2SF)	1	EOR	
2	25	3	(H,Na)ClO4	lgK:4.89(3SF)	0	MDS	140
3	25	3	NaClO4	lgK:4.09(3SF)	0	MDS	437
4	25	4	(H,Na)ClO4	lgK:5.970(0.002MD)	3	EOD	20829
5	25	4	NaClO4	lgK:5.50(3SF)	0	MIS	5077
6	25	4	NaClO4	lgK:5.97(3SF)	0	MIS	5398
7	20	4	HClO4	lgR:5.9(0.07SD)	0	CMN	437
8	20	4	HClO4	lgR:5.51(3SF)	0	MCE	437
9	20	4	HClO4	lgR:5.88(3SF)	0	MDS	437
10	20	4	HClO4	lgR:5.52(3SF)	0	MRX	437
11	20	4	HClO4	lgR:5.8(0.3MD)	0	MIS	5077
12	20	4	HClO4	lgR:5.9(0.2MD)	0	MIS	5077
13	20	4	HClO4	lgR:5.9(0.4MD)	0	MSE	13458
14	20	4	HClO4	lgR:5.89(0.02MD)	3	EOD	20829
15	20	4	HClO4	lgR:5.96(3SF)	0	MCE	22616
16	25	0.96	HClO4	lgR:5.79(3SF)	0	MCE	7998
17	25	1	HClO4	lgR:5.3(0.05SD)	0	MIX	5398
18	25	1	HClO4	lgR:5.40(0.01MD)	3	EOD	20829
19	25	1	HNO3	lgR:5.4(0.05SD)	0	MIX	5398
20	25	2	HClO4	lgR:5.81(3SF)	0	MIX	437
21	25	2	HClO4	lgR:5.80(3SF)	0	MDS	7998
22	25	2	HClO4	lgR:5.8(0.4MD)	3	EOD	20829
23	25	2	HClO4	lgR:5.81(0.02MD)	3	EOD	20829
24	25	2	HNO3	lgR:5.97(3SF)	0	MCE	5398
25	25	2.88	HClO4	lgR:5.83(3SF)	0	MCE	7033
26	25	3	HClO4	lgR:4.2(2SF)	0	ENS	437
27	27	2	HClO4	lgR:4.62(3SF)	0	MDS	437
28	25	4	HClO4	dG:-33.932(1.SD)kJ	0	MMX	4923

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29	25	4	HClO4	dH:-17.52(1.SD)kJ	0	MMX	4923

Reaction No. 2740							
Zr+4_F-1 + H+1_F-1 = Zr+4_F-1(2) + H+1							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:5.3(2SF)	1	EOR	
2	25	3	(H,Na)ClO4	lgK:3.67(3SF)	0	MDS	140
3	25	3	NaClO4	lgK:3.67(3SF)	0	MDS	437
4	25	4	(H,Na)ClO4	lgK:4.37(0.03MD)	3	EOD	20829
5	25	4	NaClO4	lgK:4.4(2SF)	0	MIS	5398
6	20	4	HClO4	lgR:4.4(0.07SD)	0	CMN	437
7	20	4	HClO4	lgR:3.7(2SF)	0	MCE	437
8	20	4	HClO4	lgR:4.54(3SF)	0	MCE	437
9	20	4	HClO4	lgR:4.36(3SF)	0	MDS	437
10	20	4	HClO4	lgR:4.04(3SF)	0	MRX	437
11	20	4	HClO4	lgR:4.3(0.1MD)	0	MIS	5077
12	20	4	HClO4	lgR:4.4(0.4MD)	0	MIS	5077
13	20	4	HClO4	lgR:4.41(3SF)	0	MCE	7033
14	20	4	HClO4	lgR:4.36(0.3MD)	0	MSE	13458
15	20	4	HClO4	lgR:3.8(0.3MD)	0	EOD	20289
16	20	4	HClO4	lgR:4.36(0.06MD)	3	EOD	20829
17	20	4	HClO4	lgR:4.41(0.16MD)	3	EOD	20829
18	20	4	HClO4	lgR:4.5(2SF)	0	MCE	22616
19	25	0.96	HClO4	lgR:4.54(3SF)	0	MCE	7033
20	25	1	HClO4	lgR:3.8(0.05SD)	0	MIX	5398
21	25	1	HNO3	lgR:4.2(0.05SD)	0	MIX	5398
22	25	2	HClO4	lgR:4.3(0.05SD)	0	MDS	140
23	25	2	HClO4	lgR:3.57(3SF)	0	MCE	437
24	25	2	HClO4	lgR:4.29(3SF)	0	MDS	7033
25	25	2	HClO4	lgR:4.32(3SF)	0	MDS	7998
26	25	2	HClO4	lgR:4.22(0.11MD)	3	EOD	20829
27	25	2	HNO3	lgR:3.59(3SF)	0	MCE	437
28	25	2.88	HClO4	lgR:4.51(3SF)	0	MCE	7033
29	25	3	HClO4	lgR:3.63(3SF)	0	ENS	437
30	25	4	HClO4	lgR:4.4(2SF)	0	MIS	5398
31	25	4	HClO4	dG:-24.665(1.SD)kJ	0	MMX	4923

32	25	4	HClO4	dH:-16.83(4SF)kJ	0	MMX	4923

Reaction No. 2741							
Zr+4_F-1(2) + H+1_F-1 = Zr+4_F-1(3) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.0(2SF)	1	ELC	
2	25	0.5	NH4ClO4	lgK:2.7(0.1MD)	0	EOD	20829
3	25	0.5	NaClO4	lgK:2.7(0.05SD)	0	MRX	140
4	25	3	(H,Na)ClO4	lgK:2.97(3SF)	0	MDS	140
5	25	3	NaClO4	lgK:2.97(3SF)	0	MDS	437
6	20	4	HClO4	lgR:3.0(0.07SD)	0	MDS	437
7	20	4	HClO4	lgR:3.04(3SF)	0	MRX	437
8	20	4	HClO4	lgR:3.00(3SF)	0	MCE	7033
9	20	4	HClO4	lgR:3.00(3SF)	0	MCE	13458
10	20	4	HClO4	lgR:3.0(0.2MD)	0	MSE	13458
11	20	4	HClO4	lgR:3.00(0.09MD)	0	EOD	20829
12	20	4	HClO4	lgR:3.14(0.05MD)	0	EOD	20829
13	25	0.5	HClO4	lgR:2.70(3SF)	0	MCE	7033
14	25	0.96	HClO4	lgR:2.96(3SF)	0	MCE	7033
15	25	2	HClO4	lgR:2.8(0.05SD)	0	MDS	5398
16	25	2	HClO4	lgR:2.88(3SF)	0	MDS	7033
17	25	2	HClO4	lgR:2.83(3SF)	0	MDS	7998
18	25	2	HClO4	lgR:3.1(0.7MD)	0	EOD	20829
19	25	2.88	HClO4	lgR:3.10(3SF)	0	MCE	7033
20	25	3	HClO4	lgR:2.21(3SF)	0	ENS	437
21	25	4	HClO4	dG:-16.898(1.SD)kJ	0	MMX	4923
22	25	4	HClO4	dH:-11.26(1.SD)kJ	0	MMX	4923

Reaction No. 2742							
Zr+4_F-1(3) + H+1_F-1 = Zr+4_F-1(4) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.4(2SF)	1	ELC	
2	25	0.5	NH4ClO4	lgK:1.73(0.07MD)	0	EOD	20829
3	25	0.5	NaClO4	lgK:1.8(0.05SD)	0	MRX	140
4	25	1	(H,Na)ClO4	lgK:2.80(0.07MD)	0	EOD	20829
5	25	3	(H,Na)ClO4	lgK:2.77(3SF)	0	MDS	140

6	25	3	NaClO4	lgK:2.77(3SF)	0	MDS	437
7	20	4	HClO4	lgR:2.3(0.07SD)	0	MDS	437
8	20	4	HClO4	lgR:2.20(3SF)	0	MRX	437
9	20	4	HClO4	lgR:2.28(3SF)	0	MCE	7033
10	20	4	HClO4	lgR:2.28(3SF)	0	MCE	13458
11	20	4	HClO4	lgR:2.2(2SF)	0	MSE	13458
12	20	4	HClO4	lgR:2.11(0.05MD)	0	EOD	20829
13	20	4	HClO4	lgR:2.28(0.13MD)	0	EOD	20829
14	25	0.5	HClO4	lgR:1.83(3SF)	0	MCE	7033
15	25	0.96	HClO4	lgR:2.44(3SF)	0	MCE	7033
16	25	2.88	HClO4	lgR:2.36(3SF)	0	MCE	7033
17	25	3	HClO4	lgR:1.94(3SF)	0	ENS	437
18	25	4	HClO4	dG:-12.61(1.SD)kJ	0	MMX	4923
19	25	4	HClO4	dH:-22.1(1.SD)kJ	0	MMX	4923

Reaction No. 2743

Zr+4_F-1(4) + H+1_F-1 = Zr+4_F-1(5) + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.4(2SF)	1	ELC	
2	25	0.5	NH4ClO4	lgK:1.51(0.07MD)	0	EOD	20829
3	25	0.5	NaClO4	lgK:1.5(0.05SD)	0	MRX	140
4	25	1	(H,Na)ClO4	lgK:1.90(0.07MD)	0	EOD	20829
5	25	3	(H,Na)ClO4	lgK:1.55(3SF)	0	MDS	140
6	25	3	NaClO4	lgK:1.55(3SF)	0	MDS	437
7	20	4	HClO4	lgR:1.5(0.07SD)	0	MDS	437
8	20	4	HClO4	lgR:1.7(2SF)	0	MRX	437
9	20	4	HClO4	lgR:1.53(0.07MD)	0	MDS	7033
10	20	4	HClO4	lgR:1.8(2SF)	0	MSE	13458
11	20	4	HClO4	lgR:1.53(0.15MD)	0	EOD	20829
12	20	4	HClO4	lgR:1.83(0.04MD)	0	EOD	20829
13	25	0.5	HClO4	lgR:1.51(3SF)	0	MCE	7033
14	25	0.96	HClO4	lgR:1.40(3SF)	0	MCE	7033
15	25	2.88	HClO4	lgR:1.56(3SF)	0	MCE	7033

Reaction No. 2744

Zr+4_F-1(5) + H+1_F-1 = Zr+4_F-1(6) + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NH4ClO4	lgK:0.85(0.04MD)	3	EOD	20829
2	25	0.5	NaClO4	lgK:0.9(0.05SD)	0	MRX	140
3	25	1	(H,Na)ClO4	lgK:1.35(0.07MD)	3	EOD	20829
4	25	3	(H,Na)ClO4	lgK:2.54(3SF)	0	MDS	140
5	25	3	NaClO4	lgK:2.54(3SF)	0	MDS	437
6	20	4	HClO4	lgR:0.3(0.07SD)	0	MDS	437
7	20	4	HClO4	lgR:0.5(1SF)	0	MRX	437
8	20	4	HClO4	lgR:0.30(2SF)	0	MCE	7033
9	20	4	HClO4	lgR:0.30(2SF)	0	MDS	7033
10	20	4	HClO4	lgR:0.3(0.4MD)	3	EOD	20829
11	20	4	HClO4	lgR:0.67(0.01MD)	3	EOD	20829
12	25	0.5	HClO4	lgR:0.86(2SF)	0	MCE	7033
13	25	0.96	HClO4	lgR:1.23(3SF)	0	MCE	7033
14	25	2.88	HClO4	lgR:1.01(3SF)	0	MCE	7033

Reaction No. 3084							
Zr+4 + NTA-3 = Zr+4_NTA-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:18.9(0.07SD)	3	MLC	4433
2	19:21	2	HClO4	lgK:18.6(3SF)	4	MIX	1118
3	20	0.23	HClO4	lgK:20.806(5SF)	6	MIX	4405
4	20	1	HClO4	lgK:19.505(5SF)	3	MIX	4405
5	20	1	NaClO4	lgK:20.2(3SF)	5	EOD	1562
6	20	2	HClO4	lgK:18.59(4SF)	5	MIX	13428
7	25	0	Inf. Dilution	lgK:24.1(3SF)	3	CRV	552
8	25	0	Inf. Dilution	lgK:20.8(3SF)	0	MGL	13427
9	25	0	Unknown	lgK:20.8(0.1MD)	0	MIX	5106
10	25	0.1	KCl	lgK:20.8(3SF)	0	CMN	1118
11	25	0	Inf. Dilution	dH:6.0(2SF)	4	CRV	552

Reaction No. 4193							
EDTA-4 + Zr+4 = Zr+4_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	2	HClO4	lgK:27.9(3SF)	0	MIX	906
2	20	0.1	Unknown	lgK:29.5(3SF)	4	ENS	773

3	20	0.23	HClO4	lgK:29.041(5SF)	3	MIX	4405
4	20	1	HClO4	lgK:28.1(3SF)	3	MIX	4405
5	20	1	HNO3	lgK:29.(0.3SD)	3	MIX	931
6	20	1	HNO3	lgK:28.46(4SF)	5	MIX	931
7	20	1	NaClO4	lgK:27.7(3SF)	4	MRX	142
8	20	1	Unknown	lgK:28.(0.4SD)	4	EIU	903
9	20	1.2	HCl	lgK:29.0(0.04SD)	3	MIX	138
10	20	2	HClO4	lgK:27.91(4SF)	0	MIX	13428
11	20	2	HNO3	lgK:30.58(4SF)	0	MIX	931
Note (11): Low wgt for internal consistency							
12	20	3	HNO3	lgK:31.11(4SF)	0	MIX	931
Note (12): Low wgt for internal consistency							
13	20	4	HNO3	lgK:30.92(4SF)	0	MIX	931
14	20	5	HNO3	lgK:30.63(4SF)	3	MIX	931
15	20	5	HNO3	lgK:30.6(0.2SD)	0	MIX	931
16	23	3.6	HNO3	lgK:28.4(3SF)	0	MPL	906
Note (16): Low wgt for internal consistency							
17	23			lgK:28.4(3SF)	0	MPL	906
18	25	0	Inf. Dilution	lgK:33.25(4SF)	1	EOR	
19	25	0	Inf. Dilution	lgK:32.8(3SF)	0	CRV	552
Note (19): Low wgt for internal consistency							
20	25	0.1	KCl	lgK:29.9(3SF)	3	MSP	140
21	25	0.1	KCl	lgK:19.9(3SF)	0	MSP	13427
22	25	0.1	KNO3	lgK:29.(0.9SD)	0	MSP	903
Note (22): Low wgt for internal consistency							
23	25	0.1	NaClO4	lgK:19.40(4SF)	0	MGL	140
24	25	0.1	Unknown	lgK:29.4(3SF)	4	ENS	773
25	25	0	Unknown	dG:-44.73(0.07MD)	2	MCL	5106
26	25	0	Inf. Dilution	dH:-0.6(1SF)	4	CRV	552
27	25	0	Unknown	dH:-0.62(0.15MD)	2	MCL	5106

Reaction No. 4194							
Zr+4_EDTA-4 + OH-1 = Zr+4_OH-1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:7.9(2SF)	4	MRX	999
2	25	0.1	Unknown	lgK:7.6(2SF)	4	ENS	773

Reaction No. 4255

2<Zr+4_EDTA-4> = Zr+4(2)_EDTA-4(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:3.5(2SF)	4	ENS	773

Reaction No. 4269

Zr+4 + DTPA-5 = Zr+4_DTPA-5

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:36.00(4SF)	3	ENS	1539
2	20	0.23	HClO4	lgK:35.806(5SF)	3	MIX	4405
3	20	0.39	HCl	lgK:34.0(3SF)	3	MIX	138
4	20	0.39	Unknown	lgK:33.96(0.02SD)	3	MSC	999
5	20	1	HClO4	lgK:33.699(5SF)	3	MIX	4405
6	20	1	NaClO4	lgK:36.9(3SF)	0	MRX	999
Note (6): Expediency (hard to tell if H+ affecting the DTPA or OH- the Zr+4)							
7	23	1	HClO4	lgK:33.69(4SF)	5	MIX	4405
8	25	0	Inf. Dilution	lgK:40.6(3SF)	2	EAE	

Reaction No. 4270

Zr+4_DTPA-5 + OH-1 = Zr+4_OH-1_DTPA-5

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:8.1(2SF)	4	MRX	999
2	20	1	NaClO4	lgK:8.1(2SF)	5	CRV	13242
3	25	0	Inf. Dilution	lgK:7.8(3SF)	2	EAE	

Reaction No. 4369

Zr+4 + H+1(2)_TTHA-6 = Zr+4_H+1(2)_TTHA-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.5	Unknown	lgK:19.7(3SF)	4	ENS	773

Reaction No. 4539

Acac-1 + Zr+4 = Zr+4_Acac-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:11.255(5SF)	6	MLC	4433
2	25	0	Inf. Dilution	lgK:11.3(3SF)	1	ESC	

Reaction No. 4907

Zr+4 + 4<e-1> = Zr(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-92.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-101.5(0.1SD)	0	ENS	5458
3	25	0	Inf. Dilution	lgK:-101.46(5SF)	0	CRV	32224
4	25	0	Inf. Dilution	dG:522.(3SF)kJ	2	CRV	7421
5	25	0	Inf. Dilution	dG:133.270(6SF)	0	CRV	9029
6	25	0	Inf. Dilution	dG:528.510(9.227MD)kJ	2	CRV	20829
7	25	0	Inf. Dilution	E:-1.45(3SF)	0	CRV	6330
Note (7): Low wgt to achieve balance							
8	25	0	Inf. Dilution	E:-1.45(3SF)	0	CRV	7761
9	25	0	Inf. Dilution	E:-1.45(3SF)	0	CRV	22519
10	25	0	Inf. Dilution	E:-1.55(3SF)	0	CRV	22551
11	25	0	Inf. Dilution	dH:636.7(4SF)kJ	0	CRV	7761
12	25	0	Inf. Dilution	dH:150.330(6SF)	1	CRV	9029
13	25	0	Inf. Dilution	dH:608.500(5.0MD)kJ	2	CRV	20829

Reaction No. 5015

Zr+4 + 2<CO3-2> + 2<OH-1> = Zr+4_OH-1(2)_CO3-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:46.9(0.1SD)	4	MSL	29848

Reaction No. 9197

Zr+4 + H+1(3)_NTA-3 = Zr+4_NTA-3 + 3<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HClO4	lgR:5.4(2SF)	3	MCE	22620
2	25	1	HClO4	lgR:5.4(0.05SD)	4	MIX	1118
3	25	2	HClO4	lgR:4.1(0.04SD)	4	MIX	1118

Reaction No. 13683

2ThenTriFAcone-1 + Zr+4 = Zr+4_2ThenTriFAcone-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:10.98(4SF)	6	MDS	999

Reaction No. 13684

Zr+4_2ThenTriFAcone-1 + 2ThenTriFAcone-1 = Zr+4_2ThenTriFAcone-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:10.90(4SF)	6	MDS	999

Reaction No. 13685

Zr+4_2ThenTriFAcone-1(2) + 2ThenTriFAcone-1 = Zr+4_2ThenTriFAcone-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:10.36(4SF)	6	MDS	999

Reaction No. 13686

Zr+4_2ThenTriFAcone-1(3) + 2ThenTriFAcone-1 = Zr+4_2ThenTriFAcone-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:9.93(3SF)	6	MDS	999

Reaction No. 14070

2<Zr+4_OH-1_EDTA-4> = Zr+4(2)_OH-1(2)_EDTA-4(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:3.5(2SF)	4	MSP	140

Reaction No. 14445

2OHTriMDTA-4 + Zr+4 = Zr+4_2OHTriMDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	HCl	lgK:23.6(0.04SD)	4	MSC	999

Reaction No. 14638

CDTA-4 + Zr+4 = Zr+4_CDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	0.1	KNO3	lgK:20.85(4SF)	5	MEF	999
2	19:21	2	HClO4	lgK:29.9(3SF)	0	MIX	999
3	20	0.1	KNO3	lgK:20.74(4SF)	5	MEF	999
4	25	0	Inf. Dilution	lgK:30.0(3SF)	1	ESC	
5	30	0.1	KNO3	lgK:20.64(4SF)	5	MEF	999

Reaction No. 18084

Zr+4_EDTA-4 + H2O = Zr+4_OH-1_EDTA-4 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	Unknown	lgK:-6.1(2SF)	6	CRV	552
2	25	0.1	KCl	lgK:-6.2(2SF)	6	MSP	999

Reaction No. 18117							
Zr+4_DTPA-5 + H2O = Zr+4_OH-1_DTPA-5 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:-5.90(3SF)	6	ENS	1539
2	25	0	Inf. Dilution	lgK:-6.31(3SF)	2	EAE	

Reaction No. 18183							
Zr+4 + H+1_Citric-3 = Zr+4_H+1_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:10.778(5SF)	2	MLC	4433
2	20	0.1	Unknown	lgK:10.80(4SF)	0	ENS	1539
3	25	0	Inf. Dilution	lgK:8.52(3SF)	2	CRV	31301

Reaction No. 18184							
Zr+4 + H+1(2)_Citric-3 = Zr+4_H+1(2)_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:6.20(3SF)	6	ENS	1539
2	23	2	HClO4	lgK:3.41(3SF)	5	MCE	3262
3	25	0	Inf. Dilution	lgK:7.16(3SF)	2	EAE	

Reaction No. 18266							
Oxalic-2 + Zr+4 = Zr+4_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:10.26(0.05SD)	2	MLC	4433
2	20	0.1	Unknown	lgK:10.40(4SF)	6	ENS	1539
3	20	1	HNO3	lgK:11.1(0.2SD)	2	MIX	931
4	20	5	HNO3	lgK:11.3(0.2SD)	3	MIX	931
5	23	2	HClO4	lgK:5.50(3SF)	0	MCE	3262
6	23	2	HClO4	lgK:5.0(2SF)	0	MCE	7744
7	25	0	Inf. Dilution	lgK:12.11(4SF)	1	EOR	
8	25	0	Inf. Dilution	lgK:12.1(3SF)	2	EAE	
9	25	0	Inf. Dilution	lgK:10.52(4SF)	0	CRV	31301

Reaction No. 18267							
2<Oxalic-2> + Zr+4 = Zr+4_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:19.40(4SF)	6	ENS	1539

2	20	1	KNO3	lgK:31.40(4SF)	5	MIX	11516
Note (2): Caletka R et al., J. Inorg. Nucl. Chem., 1964, 26, 1443							
3	25	0	Inf. Dilution	lgK:20.42(4SF)	1	EOR	
4	25	0	Inf. Dilution	lgK:20.4(3SF)	2	EAE	
5	25	0	Inf. Dilution	lgK:18.15(4SF)	0	CRV	31301

Reaction No. 18327

Zr+4 + H+1_Tartaric-2 = Zr+4_H+1_Tartaric-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:6.9(0.07SD)	4	MLC	4433
2	20	0.1	Unknown	lgK:6.00(3SF)	6	ENS	1539
3	23	2	HClO4	lgK:2.49(3SF)	0	MCE	3262

Reaction No. 18328

Tiron-4 + Zr+4 = Zr+4_Tiron-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:24.16(4SF)	6	ENS	1539

Reaction No. 18523

Zr+4 + H+1_Gly-1 = Zr+4_H+1_Gly-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:4.0(0.09SD)	4	MGL	1555

Reaction No. 19828

Zr+4_Oxalic-2 + Oxalic-2 = Zr+4_Oxalic-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HNO3	lgK:20.3(0.3SD)	0	MIX	931
2	23	2	HClO4	lgK:9.68(3SF)	0	MCE	3262
3	25	0	Inf. Dilution	lgK:8.3(2SF)	1	ELC	

Reaction No. 20128

Zr+4_DHEGly-1_OH-1 + H+1 = Zr+4_DHEGly-1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:3.9(2SF)	4	MGL	999

Reaction No. 20129

Zr+4_DHEGly-1_OH-1(2) + H+1 = Zr+4_DHEGly-1_OH-1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:8.5(2SF)	4	MGL	999

Reaction No. 20759

2<2ThenTriFAcone-1> + Zr+4 = Zr+4_2ThenTriFAcone-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:21.88(4SF)	6	ENS	773

Reaction No. 20760

3<2ThenTriFAcone-1> + Zr+4 = Zr+4_2ThenTriFAcone-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:32.24(4SF)	6	ENS	773

Reaction No. 20761

4<2ThenTriFAcone-1> + Zr+4 = Zr+4_2ThenTriFAcone-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:42.17(4SF)	6	ENS	773

Reaction No. 23024

2<Zr+4_EDTA-4_Tiron-4> + 2<H+1> = 2<Zr+4_H+1_EDTA-4_Tiron-4>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:3.70(3SF)	4	MGL	999

Reaction No. 24037

EEDTA-4 + Zr+4 = Zr+4_EEDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:24.72(4SF)	6	MCG	999

Reaction No. 24057

TEDTA-4 + Zr+4 = Zr+4_TEDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:23.17(4SF)	6	MCG	999

Reaction No. 24364

HOEDTA-3 + Zr+2 = Zr+2_HOEDTA-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:14.42(4SF)	6	MGL	145

Reaction No. 29171							
Zr+4_DTPA-5 + H+1 = Zr+4_H+1_DTPA-5							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:1.5(2SF)	3	ENS	2373
2	25	0	Inf. Dilution	lgK:1.71(3SF)	2	EAE	

Reaction No. 32312							
Zr+4_OH-1 + H+1_Mucic-2 = Zr+4_Mucic-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:9.49(3SF)	5	MSP	2261

Reaction No. 34302							
Zr+4_OH-1(2) + H+1_Chromotropic-4 = Zr+4_OH-1_Chromotropic-4 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	16:23	0.1	KCl	lgK:18.68(4SF)	5	MSP	931

Reaction No. 34591							
6<F-1> + Zr+4 = Zr+4_F-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	Unknown	lgK:38.4(3SF)	0	CRV	2556
2	25	0	Inf. Dilution	lgK:38.110(0.430MD)	3	CRV	20829
3	25	3	HClO4	lgK:12.62(4SF)	0	ENS	437

Reaction No. 37309							
Salicylaldoxime-2 + Zr+4 = Zr+4_Salicylaldoxime-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.9(3SF)	0	MGL	931
Note (1): Low wgt for internal consistency							
2	25	0.01	KCl	lgK:17.35(4SF)	0	MGL	931
3	25	0.025	KCl	lgK:16.45(4SF)	0	MGL	931
4	25	0.05	KCl	lgK:15.13(4SF)	0	MGL	931
5	25	0.075	KCl	lgK:13.90(4SF)	0	MGL	931
6	25	0.1	KCl	lgK:12.43(4SF)	3	MGL	931

Reaction No. 37463							
Benzoylacetone-1 + Zr+4 = Zr+4_Benzoylacetone-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	1	Benzene Unknown	lgK:12.71(0.04SD)	5	MDS	5076

Reaction No. 37464

Zr+4_Benzoylacetone-1 + Benzoylacetone-1 = Zr+4_Benzoylacetone-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Benzene Unknown	lgK:11.86(0.02SD)	5	MDS	5076

Reaction No. 37465

Zr+4_Benzoylacetone-1(2) + Benzoylacetone-1 = Zr+4_Benzoylacetone-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Benzene Unknown	lgK:11.34(0.04SD)	5	MDS	5076

Reaction No. 37466

Zr+4_Benzoylacetone-1(3) + Benzoylacetone-1 = Zr+4_Benzoylacetone-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Benzene Unknown	lgK:11.08(0.04SD)	5	MDS	5076

Reaction No. 38777

Cys-2 + Zr+4 = Zr+4_Cys-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.1	KNO3	lgK:14.40(4SF)	0	MGL	905
2	30	0.1	KNO3	lgK:14.15(4SF)	0	MGL	905

Reaction No. 39154

BenzHydroxam-1 + Zr+4 = Zr+4_BenzHydroxam-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:12.43(4SF)	5	MDS	931
2	25	1	HClO4	lgK:12.431(5SF)	5	MPT	4210

Reaction No. 39155

2<BenzHydroxam-1> + Zr+4 = Zr+4_BenzHydroxam-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:24.08(4SF)	5	MDS	931
2	25	1	HClO4	lgK:24.079(5SF)	5	MPT	4210

Reaction No. 40327

MeDiPhos-4 + ZrO+2 = ZrO+2_MeDiPhos-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:13.13(4SF)	5	MGL	931

Reaction No. 40328

2<MeDiPhos-4> + ZrO+2 = ZrO+2_MeDiPhos-4(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:19.45(4SF)	5	MGL	931

Reaction No. 40329

ZrO+2 + H+1_MeDiPhos-4 = ZrO+2_H+1_MeDiPhos-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:9.01(3SF)	5	MGL	931

Reaction No. 40330

ZrO+2 + 2<H+1_MeDiPhos-4> = ZrO+2_H+1(2)_MeDiPhos-4(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:12.18(4SF)	5	MGL	931

Reaction No. 40409

Zr+4_Oxalic-2(2) + Oxalic-2 = Zr+4_Oxalic-2(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.0(2SF)	1	ESC	
2	25			lgK:4.0(0.1SD)	0	MGL	931

Reaction No. 40410

Zr+4 + H+1(2)_Oxalic-2 = Zr+4_Oxalic-2 + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.581(4SF)	1	ELC	
2	23	2	HClO4	lgR:5.5(0.04SD)	3	MIX	931
3	23	4	HClO4	lgR:5.6(0.07SD)	3	MIX	931

Reaction No. 40917

Zr+4 + H+1(2)_234TriOHPentDioic-2 = Zr+4_H+1_234TriOHPentDioic-2 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgR:3.09(3SF)	5	MIX	931

2	23	2	HClO4	lgR:3.41(0.01SD)	5	MIX	3262

Reaction No. 40918

Zr+4 + 2<H+1(2)_234TriOHPentDioic-2> = Zr+4_H+1(2)_234TriOHPentDioic-2(2) + 2<H+1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgR:4.14(3SF)	3	MIX	931
2	23	2	HClO4	lgR:5.4(0.07SD)	3	MIX	3262

Reaction No. 41690

Zr+4_OH-1(3) + H+1(3)_Pyrogallol-3 = Zr+4_H+1_OH-1(2)_Pyrogallol-3 + H+1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0	Inf. Dilution	lgK:4.17(3SF)	4	MSP	4406
2	18:20	0.1	KCl	lgK:4.06(3SF)	4	MSP	4406

Reaction No. 41727

Mandelic-1 + Zr+4 = Zr+4_Mandelic-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:5.6(0.07SD)	5	MLC	4433

Reaction No. 41936

Zr+4_EDTA-4 + H+1(2)_Tiron-4 = Zr+4_H+1_EDTA-4_Tiron-4 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:-0.92(2SF)	3	MGL	13427
2	25	0.1	Unknown	lgK:-12.(2SF)	0	MGL	140

Reaction No. 42068

SulfSal-3 + Zr+4 = Zr+4_SulfSal-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	HCl	lgK:4.792(4SF)	0	MCL	4433
2	25	0	Inf. Dilution	lgK:4.8(2SF)	1	ESC	

Reaction No. 42252

Zr+4 + H+1(2)_Lactic-1 = Zr+4_H+1_Lactic-1 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.28(3SF)	2	EAE	
2	23	2	HClO4	lgR:2.28(3SF)	5	MIX	3262

Reaction No. 42253

Zr+4 + 2<H+1(2)_Lactic-1> = Zr+4_H+1(2)_Lactic-1(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.54(3SF)	2	EAE	
2	23	2	HClO4	lgR:2.54(3SF)	5	MIX	3262

Reaction No. 42255							
Zr+4 + H+1(2)_Malic-2 = Zr+4_H+1_Malic-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.2399(4SF)	1	ELC	
2	23	2	HClO4	lgR:2.24(3SF)	3	MIX	3262

Reaction No. 42257							
Zr+4 + H+1(2)_Tartaric-2 = Zr+4_H+1_Tartaric-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgR:2.49(3SF)	5	MIX	3262

Reaction No. 42259							
Zr+4 + H+1(2)_Citric-3 = Zr+4_H+1_Citric-3 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.75(3SF)	1	ELC	
2	23	2	HClO4	lgR:3.42(3SF)	3	MIX	3262

Reaction No. 42375							
Zr+4 + H+1(2)_Oxalic-2 = Zr+4_H+1_Oxalic-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.23(3SF)	2	EAE	
2	23	2	HClO4	lgR:4.50(3SF)	4	MIX	3262

Reaction No. 42376							
Zr+4 + 2<H+1(2)_Oxalic-2> = Zr+4_H+1(2)_Oxalic-2(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.53(3SF)	2	EAE	
2	23	2	HClO4	lgR:8.32(3SF)	4	MIX	3262

Reaction No. 42378							
Zr+4 + 4<H+1(2)_Oxalic-2> = Zr+4_H+1(4)_Oxalic-2(4) + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:5.85(3SF)	2	EAE	
2	23	2	HClO4	lgR:4.397(4SF)	4	MIX	3262

Reaction No. 42783

Zr+4_OH-1(3) + Formic-1 = Zr+4_Formic-1_OH-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:3.4(0.04SD)	4	MHY	906

Reaction No. 42941

Zr+4_OH-1(3) + Acetic-1 = Zr+4_Acetic-1_OH-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:3.35(3SF)	4	MHY	906

Reaction No. 42942

Zr+4_Acetic-1_OH-1(3) + Acetic-1 = Zr+4_Acetic-1(2)_OH-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:1.83(3SF)	4	MHY	906

Reaction No. 42943

Zr+4_OH-1(2) + Acetic-1 = Zr+4_Acetic-1_OH-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.2(0.05SD)	5	MKN	906

Reaction No. 42952

Zr+4_OH-1(2) + Glycolic-1 = Zr+4_Glycolic-1_OH-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:6.5(0.03SD)	5	MSP	906
2	25	0	Inf. Dilution	lgK:7.7(2SF)	5	MKN	906

Reaction No. 42953

Zr+4_OH-1(3) + Glycolic-1 = Zr+4_Glycolic-1_OH-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:6.6(0.03SD)	5	MSP	906

Reaction No. 43055

Zr+4_OH-1(3) + Propanoic-1 = Zr+4_OH-1(3)_Propanoic-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:3.8(0.07SD)	5	MHY	906

Reaction No. 43270							
Zr+4_OH-1(3) + Tartaric-2 = Zr+4_OH-1(3)_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15			lgK:6.24(3SF)	4	MHY	906
2	20			lgK:6.15(3SF)	4	MHY	906
3	25			lgK:6.07(3SF)	4	MHY	906
4	25			lgK:6.09(3SF)	4	MHY	906
5	30			lgK:5.92(3SF)	4	MHY	906
6	35			lgK:5.84(3SF)	4	MHY	906

Reaction No. 43271							
Zr+4_OH-1(3)_Tartaric-2 + Tartaric-2 = Zr+4_OH-1(3)_Tartaric-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:3.06(3SF)	4	MHY	906

Reaction No. 43272							
Zr+4_OH-1(2) + H+1(2)_Tartaric-2 = Zr+4_H+1(2)_OH-1(2)_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:9.8(0.03SD)	4	MIX	906

Reaction No. 43276							
Zr+4_OH-1(3) + Butanoic-1 = Zr+4_Butanoic-1_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:3.8(0.06SD)	4	MHY	906

Reaction No. 43356							
Zr+4 + Asp-2 = Zr+4_Asp-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:9.70(3SF)	5	MGL	22615

Reaction No. 43357							
Zr+4_Asp-2 + Asp-2 = Zr+4_Asp-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.85(3SF)	4	MGL	22615

Reaction No. 43358
Zr+4_Asp-2(2) + Asp-2 = Zr+4_Asp-2(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:3.50(3SF)	4	MGL	22615

Reaction No. 43425							
Asn-1 + Zr+4 = Zr+4_Asn-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:8.80(3SF)	4	MGL	906

Reaction No. 43426							
Zr+4_Asn-1 + Asn-1 = Zr+4_Asn-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.25(3SF)	4	MGL	906

Reaction No. 43584							
Zr+4_OH-1(3) + 234TriOHPentDioic-2 = Zr+4_OH-1(3)_234TriOHPentDioic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:6.86(3SF)	4	MHY	906

Reaction No. 43585							
Zr+4_OH-1(2) + 234TriOHPentDioic-2 = Zr+4_OH-1(2)_234TriOHPentDioic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2.25	NaCl	lgK:11.4(0.1SD)	4	MIX	906

Reaction No. 43589							
Zr+4_OH-1(3) + Valeric-1 = Zr+4_OH-1(3)_Valeric-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:3.9(0.09SD)	4	MHY	906

Reaction No. 43628							
Glutaric-2 + Zr+4 = Zr+4_Glutaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:9.60(3SF)	4	MGL	906

Reaction No. 43629							
Zr+4_Glutaric-2 + Glutaric-2 = Zr+4_Glutaric-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.40(3SF)	4	MGL	906

Reaction No. 43630							
Zr+4_Glutaric-2(2) + Glutaric-2 = Zr+4_Glutaric-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:3.32(3SF)	4	MGL	906

Reaction No. 43639							
Gln-1 + Zr+4 = Zr+4_Gln-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:8.75(3SF)	4	MGL	906

Reaction No. 43640							
Zr+4_Gln-1 + Gln-1 = Zr+4_Gln-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.10(3SF)	4	MGL	906

Reaction No. 43705							
Zr+4 + H+1_Ascorbic-2 = Zr+4_H+1_Ascorbic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:9.5(2SF)	4	MSP	906

Reaction No. 43768							
Zr+4_OH-1(3) + Mucic-2 = Zr+4_OH-1(3)_Mucic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:6.55(3SF)	4	MHY	906

Reaction No. 43776							
2<Gluconic-1> + ZrO+2 = ZrO+2_Gluconic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0.2	NaClO4	lgK:2.25(3SF)	4	MSP	906

Reaction No. 43777							
Zr+2_OH-1(2) + Gluconic-1 = Zr+2_Gluconic-1_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HCl	lgK:6.60(3SF)	4	MIX	906
2	23	2	NaCl	lgK:6.60(3SF)	4	MIX	906

Reaction No. 43778
Zr+2_Gluconic-1_OH-1(2) + Gluconic-1 = Zr+2_Gluconic-1(2)_OH-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HCl	lgK:5.95(3SF)	4	MIX	906
2	23	2	NaCl	lgK:5.95(3SF)	4	MIX	906

Reaction No. 43986

Zr+4_OH-1(4) + H+1_Cupferron-1 = Zr+4_Cupferron-1_OH-1(3) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Vol	lgK:7.36(3SF)	4	MSP	906

Reaction No. 43987

Zr+4_Cupferron-1_OH-1(3) + H+1_Cupferron-1 = Zr+4_Cupferron-1(2)_OH-1(2) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Vol	lgK:6.74(3SF)	4	MSP	906

Reaction No. 43988

Zr+4_Cupferron-1(2)_OH-1(2) + H+1_Cupferron-1 = Zr+4_Cupferron-1(3)_OH-1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Vol	lgK:6.21(3SF)	4	MSP	906

Reaction No. 43989

Zr+4_Cupferron-1(3)_OH-1 + H+1_Cupferron-1 = Zr+4_Cupferron-1(4) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		Ethanol:Water 40:60Vol	lgK:5.84(3SF)	4	MSP	906

Reaction No. 44022

2<NTA-3> + Zr+4 = Zr+4_NTA-3(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:7.8(0.2SD)	0	MSP	906
2	25	0	Inf. Dilution	lgK:32.5(3SF)	1	ELC	

Reaction No. 44202

Zr+4_OH-1(3) + H+1_Salicylic-2 = Zr+4_H+1_OH-1(3)_Salicylic-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:6.38(0.01SD)	4	MSP	906
2	25			lgK:6.1(0.2SD)	4	MKN	906

Reaction No. 44203

Zr+4_OH-1(2) + H+1_Salicylic-2 = Zr+4_H+1_OH-1(2)_Salicylic-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:6.0(0.07SD)	4	MSP	906

Reaction No. 44344

Zr+4_OH-1(3) + H+1_SulfSal-3 = Zr+4_H+1_OH-1(3)_SulfSal-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HCl	lgK:6.2(0.2SD)	4	MSP	906
2	23	1	KCl	lgK:6.2(0.2SD)	4	MSP	906
3	25	0	Inf. Dilution	lgK:6.2(2SF)	1	ESC	
4	25			lgK:6.0(0.2SD)	0	MKN	906

Reaction No. 44345

Zr+4_OH-1(2) + H+1_SulfSal-3 = Zr+4_H+1_OH-1(2)_SulfSal-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HCl	lgK:6.2(0.06SD)	4	MSP	906
2	23	1	KCl	lgK:6.2(0.06SD)	4	MSP	906
3	25	0	Inf. Dilution	lgK:6.2(2SF)	1	ESC	

Reaction No. 44455

AmPhenMeDiPhos-4 + ZrO+2 = ZrO+2_AmPhenMeDiPhos-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:17.08(4SF)	5	MGL	906

Reaction No. 44456

2<AmPhenMeDiPhos-4> + ZrO+2 = ZrO+2_AmPhenMeDiPhos-4(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:21.66(4SF)	5	MGL	906

Reaction No. 44457

ZrO+2 + H+1_AmPhenMeDiPhos-4 = ZrO+2_H+1_AmPhenMeDiPhos-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:11.61(4SF)	5	MGL	906

Reaction No. 44458

2<ZrO+2> + AmPhenMeDiPhos-4 = ZrO+2(2)_AmPhenMeDiPhos-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:23.18(4SF)	5	MGL	906

Reaction No. 44459

ZrO+2 + H+1 + 2<AmPhenMeDiPhos-4> = ZrO+2_H+1_AmPhenMeDiPhos-4(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:16.69(4SF)	5	MGL	906

Reaction No. 44469

Zr+4_OH-1(2) + 2<PhenAcet-1> = Zr+4_OH-1(2)_PhenAcet-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.6(2SF)	4	MKN	906
2	25	0.01	Unknown	lgK:6.8(2SF)	4	MKN	906
3	25	0.1	Unknown	lgK:6.2(2SF)	4	MKN	906

Reaction No. 44486

Zr+4_OH-1(2) + Mandelic-1 = Zr+4_Mandelic-1_OH-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:6.7(0.04SD)	4	MSP	906

Reaction No. 44487

Zr+4_OH-1(3) + Mandelic-1 = Zr+4_Mandelic-1_OH-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KCl	lgK:6.71(0.02SD)	4	MSP	906

Reaction No. 44739

Zr+4_OH-1(2) + Bipy = Zr+4_Bipy_OH-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:9.7(0.03SD)	4	MSP	906

Reaction No. 44782

F*3Benzoylacetone-1 + Zr+4 = Zr+4_F*3Benzoylacetone-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:10.85(4SF)	4	MDS	906

Reaction No. 44783

Zr+4_F*3Benzoylacetone-1 + F*3Benzoylacetone-1 = Zr+4_F*3Benzoylacetone-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	1	Unknown	lgK:10.35(4SF)	4	MDS	906

Reaction No. 44784

Zr+4_F*3Benzoylacetone-1(2) + F*3Benzoylacetone-1 = Zr+4_F*3Benzoylacetone-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:9.85(3SF)	4	MDS	906

Reaction No. 44785

Zr+4_F*3Benzoylacetone-1(3) + F*3Benzoylacetone-1 = Zr+4_F*3Benzoylacetone-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:9.35(3SF)	4	MDS	906

Reaction No. 44903

Zr+4_EDTA-4 + EDTA-4 = Zr+4_EDTA-4(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:8.(0.3SD)	4	MSP	906
2	25	0	Inf. Dilution	lgK:8.0(2SF)	1	ESC	

Reaction No. 44913

EDDS-4 + ZrO+2 = ZrO+2_EDDS-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	30	0.1	KNO3	lgK:11.20(4SF)	4	MGL	906

Reaction No. 45069

Zr+4_OH-1(3) + H+1_PAR-2 = Zr+4_H+1_OH-1(3)_PAR-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:16.44(4SF)	4	MSP	906

Reaction No. 45087

TriMDTA-4 + Zr+4 = Zr+4_TriMDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HClO4	lgK:28.33(4SF)	4	MDS	906

Reaction No. 45088

Zr+4 + 2<H+1_TriMDTA-4> = Zr+4_H+1(2)_TriMDTA-4(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HClO4	lgK:33.02(4SF)	4	MDS	906

Reaction No. 45111							
Zr+4_OH-1(2) + 2<Phenanth> = Zr+4_Phenanth(2)_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1:1	Unknown	lgK:10.07(0.01SD)	4	MSP	906

Reaction No. 45131							
EDDG-4 + ZrO+2 = ZrO+2_EDDG-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	30	0.1	KNO3	lgK:11.80(4SF)	4	MGL	906

Reaction No. 45283							
Zr+4 + H+1(5)_DTPA-5 = Zr+4_DTPA-5 + 5<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:4.43(3SF)	0	MSP	906
2	25	0	Inf. Dilution	lgK:9.9(2SF)	1	ELC	

Reaction No. 46382							
Zr+4_Oxalic-2 + H+1 + H2O = Zr+4_OH-1 + H+1(2)_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.265(4SF)	1	ELC	
2	25	4	NaClO4	lgK:4.30(3SF)	0	MDS	4708

Reaction No. 52453							
Zr+4_Cl-1(4) + 2<2NitrAniline> = Zr+4_2NitrAniline(2)_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		Dioxan	lgK:4.78(3SF)	4	MSP	4695

Reaction No. 52454							
Zr+4_Cl-1(4) + 2<35DiNitrAniline> = Zr+4_35DiNitrAniline(2)_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		Dioxan	lgK:4.15(3SF)	4	MSP	4695

Reaction No. 52455							
Zr+4_Cl-1(4) + 2<2Me4NitrAniline> = Zr+4_2Me4NitrAniline(2)_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		Dioxan	lgK:5.58(3SF)	4	MSP	4695

Reaction No. 52994

Cl-1 + Zr+4 = Zr+4_Cl-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:30	2	HClO4	lgK:0.03(1SF)	6	CRV	20829
2	18:30	2	HClO4	lgK:-0.02(1SF)	4	MIX	22619
3	18:30	4	HClO4	lgK:0.05(1SF)	6	CRV	20829
4	18:30	4	HClO4	lgK:0.04(1SF)	4	MIX	22619
5	20	2	HClO4	lgK:0.08(1SF)	5	MDS	5398
6	20	2	HClO4	lgK:0.079(2SF)	4	MIX	13428
7	20	3.5	HClO4	lgK:-0.5(1SF)	0	MSP	5398
8	20	6.54	H(ClO4)	lgK:0.92(2SF)	3	MDS	140
9	23	2	HClO4	lgK:-0.02(1SF)	3	MIX	3262
10	23	4	H(ClO4)	lgK:0.04(1SF)	3	MCE	140
11	23	4	HClO4	lgK:0.04(1SF)	3	MIX	3262
12	25	0	Inf. Dilution	lgK:0.3(0.05SD)	0	ENS	5458
13	25	0	Inf. Dilution	lgK:0.19(0.11SD)	2	EOD	7033
14	25	0	Inf. Dilution	lgK:1.590(0.060MD)	6	CRV	20829
15	25	0	Inf. Dilution	lgK:0.50(2SF)	4	CRV	32224
16	25	2	HClO4	lgK:0.30(2SF)	3	MDS	140
17	25	2	HClO4	lgK:0.30(2SF)	2	MSE	7998
18	25	2	HClO4	lgK:-0.02(1SF)	5	MIX	22619
19	25	6.5	HClO4	lgK:0.92(2SF)	4	MIX	5502
20	25	6.54	HCl	lgR:0.92(0.05SD)	5	MDS	5502

Reaction No. 52995

Zr+4 + 2<Cl-1> = Zr+4_Cl-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:-0.54(2SF)	0	MDS	5398
2	23	4	H(ClO4)	lgK:-0.68(2SF)	1	MCE	140
3	25	0	Inf. Dilution	lgK:0.48(2SF)	1	ELC	
4	25	0	Inf. Dilution	lgK:0.48(2SF)	1	EOR	
5	25	0	Inf. Dilution	lgK:2.170(0.240MD)	0	CRV	20829
6	25	6.54	HCl	lgR:1.32(0.05SD)	2	MDS	5502

Reaction No. 52996

Zr+4 + 3<Cl-1> = Zr+4_Cl-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	20	2	HClO4	lgK:-1.0(2SF)	0	MDS	5398
2	23	4	H(ClO4)	lgK:-1.30(3SF)	0	MCE	140
3	25	0	Inf. Dilution	lgK:-0.6(1SF)	1	ELC	
4	25	6.54	HCl	lgR:1.51(0.05SD)	0	MDS	5502

Reaction No. 53080							
5<CO3-2> + Zr+4 = Zr+4_CO3-2(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.6(0.2SD)	4	MSL	29848
2	25	0	Inf. Dilution	lgK:43.(2SF)	2	EOD	29848

Reaction No. 53160							
Zr+4_F-1(5) + F-1 = Zr+4_F-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.5	HCl	lgK:3.6(2SF)	0	MEF	5398
2	23			lgK:3.84(3SF)	0	MIX	5398
3	25	0	Inf. Dilution	lgK:3.3(3SF)	0	EAE	
4	25	0	Inf. Dilution	lgK:0.75(0.38SD)	0	EOD	7033
5	25	1	NaClO4	lgK:1.4(0.05SD)	0	MQH	931
Note (5): Low wgt for internal consistency (not in accord with no-I data)							
6	25	0	Inf. Dilution	dH:4.0(5.SD)	0	MCL	437
7	25			dH:3.800(4SF)	0	MCL	437

Reaction No. 53161							
Zr+4_F-1(2) + 2<H+1> = Zr+4 + 2<H+1_F-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.7(2SF)	0	ELC	
2	25	1	HClO4	lgR:-9.11(3SF)	0	MCE	5398
3	25	1	HNO3	lgR:-9.65(3SF)	0	MCE	5398
4	25	2	HClO4	lgR:-9.38(3SF)	0	MCE	5398
5	25	2	HNO3	lgR:-9.53(3SF)	0	MCE	5398

Reaction No. 53302							
Zr+4 + H+1_SO4-2 = Zr+4_SO4-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.6(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:2.6(2SF)	0	EDH	29883

Note (2): K > 4.6E2							
3	20	2	HClO4	lgR:2.6(2SF)	5	MDS	5398
4	20	2	HClO4	lgR:2.6(2SF)	4	MIX	13428
5	20	4	HClO4	lgR:2.85(3SF)	3	MCE	140
6	20	4	HClO4	lgR:2.7(2SF)	3	MSE	13458
7	20	4	HClO4	lgR:2.8(2SF)	3	MCE	22616
8	23	2.3	HClO4	lgR:2.56(3SF)	4	MCE	140
9	23	2.3	HClO4	lgR:2.67(3SF)	5	MCE	931
10	25	2	HClO4	lgR:2.66(3SF)	5	MDS	140

Reaction No. 53303							
Zr+4_SO4-2 + H+1_SO4-2 = Zr+4_SO4-2(2) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.2(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1.72(3SF)	0	EDH	29883
3	20	2	HClO4	lgR:1.9(2SF)	4	MIX	13428
4	20	4	HClO4	lgR:1.85(3SF)	5	MCE	140
5	20	4	HClO4	lgR:0.9(1SF)	3	MDS	5398
6	20	4	HClO4	lgR:1.9(2SF)	4	MSE	13458
7	20	4	HClO4	lgR:1.85(3SF)	4	MCE	22616
8	25	2	HClO4	lgR:1.72(3SF)	5	MDS	140

Reaction No. 53304							
Zr+4_SO4-2(2) + H+1_SO4-2 = Zr+4_SO4-2(3) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.24(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-0.2(1SF)	0	ELC	
3	25	0	Inf. Dilution	lgK:0.0(1SF)	0	EDH	29883
4	20	2	HClO4	lgR:1.0(2SF)	2	MIX	13428
5	20	4	HClO4	lgR:0.3(1SF)	3	MDS	5398
6	25	2	HClO4	lgR:0.0(1SF)	3	MDS	140

Reaction No. 53305							
Zr+4 + 2<H+1_SO4-2> = Zr+4_SO4-2(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.7(2SF)	1	ELC	
2	20	2	HClO4	lgR:4.5(2SF)	3	MDS	5398

Note (2): Low wgt for internal consistency							
3	23	2.3	HClO4	lgR:3.34(3SF)	3	MCE	140
4	23	2.3	HClO4	lgR:3.54(3SF)	3	MCE	931

Reaction No. 53306

Zr+4 + 3<H+1_SO4-2> = Zr+4_SO4-2(3) + 3<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.4(2SF)	1	ELC	
2	20	2	HClO4	lgR:5.5(2SF)	3	MDS	5398
3	23	2.3	HClO4	lgR:5.61(3SF)	3	MCE	140
4	23	2.3	HClO4	lgR:5.59(3SF)	3	MCE	931

Reaction No. 53370

Zr+4 + 3<H+1_PO4-3> = Zr+4_H+1(3)_PO4-3(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:1.59(3SF)	5	MSP	5398
2	25	0	Inf. Dilution	lgK:5.14(3SF)	2	EAE	

Reaction No. 53503

Zr+4 + H2O2 = Zr+4_H2O2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HCl	lgK:3.10(3SF)	3	MSP	5398
2	25	0	Inf. Dilution	lgK:3.1(3SF)	2	EAE	

Reaction No. 53504

ZrO+2 + H2O2 = ZrO+2_H2O2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:4.0(2SF)	0	MDG	5398

Reaction No. 53656

Zr+4 + NO3-1 = Zr+4_NO3-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:30	2	HClO4	lgK:0.05(1SF)	6	CRV	20829
2	18:30	2	HClO4	lgK:-0.03(1SF)	4	MIX	22619
3	18:30	4	HClO4	lgK:-0.09(1SF)	6	CRV	20829
4	18:30	4	HClO4	lgK:-0.04(1SF)	4	MIX	22619
5	20	2	HClO4	lgK:-0.05(1SF)	4	MDS	5398

6	20	2	HClO4	lgK:-0.04(1SF)	4	MIX	13428
7	20	4	HClO4	lgK:0.34(2SF)	3	MDS	140
8	23	2	HClO4	lgK:-0.04(1SF)	4	MCE	140
9	23	2	HClO4	lgK:-0.03(1SF)	3	MIX	3262
10	23	4	HClO4	lgK:-0.06(1SF)	4	MCE	140
11	23	4	HClO4	lgK:-0.06(1SF)	3	MIX	3262
12	25	0	Inf. Dilution	lgK:0.3(0.05SD)	0	ENS	5458
13	25	0	Inf. Dilution	lgK:1.590(0.080MD)	6	CRV	20829
14	25	0	Inf. Dilution	lgK:0.3(1SF)	0	CRV	32224
15	25	2	HClO4	lgK:0.3(1SF)	2	MDS	140
16	25	2	HClO4	lgK:0.30(2SF)	2	MSE	7998
17	25	2	HClO4	lgK:-0.05(1SF)	5	MIX	13428
18	25	4	HClO4	lgK:0.34(2SF)	2	MIX	5502
19	25	4	HNO3	lgR:0.34(0.05SD)	5	MDS	5502

Reaction No. 53657

Zr+4 + 2<NO3-1> = Zr+4_NO3-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:-0.46(2SF)	2	MDS	5398
2	23	2	HClO4	lgK:-0.34(2SF)	0	MCE	140
3	23	2	HClO4	lgK:0.22(2SF)	0	MCE	140
4	23	4	HClO4	lgK:-0.85(2SF)	0	MCE	140
5	25	0	Inf. Dilution	lgK:3.3(2SF)	1	ELC	
6	25	0	Inf. Dilution	lgK:2.640(0.170MD)	0	CRV	20829
7	25	4	HNO3	lgR:0.11(0.05SD)	3	MDS	5502

Reaction No. 54492

F-1 + Zr+4 = Zr+4_F-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.4(2SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:9.8(0.05SD)	0	ENS	5458
3	25	0	Inf. Dilution	lgK:5.84(0.07SD)	0	EOD	7033
4	25	0	Inf. Dilution	lgK:10.120(0.070MD)	3	CRV	20829
5	25	0	Inf. Dilution	lgK:9.8(2SF)	0	CRV	32224
6	25	2	Unknown	lgK:8.94(3SF)	0	CRV	2556
7	25	4	Unknown	lgK:9.44(3SF)	0	CRV	2556

8	25	4	HClO4	dG:-54.1(3SF)kJ	0	MMX	4923
9	25	0	Inf. Dilution	dH:-17.500(0.700MD)kJ	3	CRV	20829
10	25	4	HClO4	dH:-5.6(2SF)kJ	0	MMX	4923
11	25	4	Unknown	dH:-1.3(2SF)	0	CRV	2556
12	25			dH:5.975(4SF)	0	MCL	437

Reaction No. 54493

Zr+4_F-1 + F-1 = Zr+4_F-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:4.39(0.15SD)	0	EOD	7033
3	25	4	HClO4	dG:-44.9(3SF)kJ	2	MMX	4923
4	25	4	HClO4	dH:-4.9(2SF)kJ	2	MMX	4923
5	25			dH:5.975(4SF)	0	MCL	437

Reaction No. 54494

Zr+4_F-1(2) + F-1 = Zr+4_F-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.1(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:2.89(0.16SD)	0	EOD	7033
3	25	4	HClO4	dG:-37.1(3SF)kJ	0	MMX	4923
4	25	4	HClO4	dH:0.7(1SF)kJ	0	MMX	4923
5	25			dH:5.975(4SF)	0	MCL	437

Reaction No. 54495

Zr+4_F-1(3) + F-1 = Zr+4_F-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.7(2SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:2.18(0.26SD)	0	EOD	7033
3	25	1	NaClO4	lgK:2.8(0.05SD)	0	MQH	931
4	25	4	HClO4	dG:-32.8(3SF)kJ	0	MMX	4923
5	25	0	Inf. Dilution	dH:10.(5.SD)	0	MCL	437
6	25	4	HClO4	dH:-10.(2SF)kJ	0	MMX	4923
7	25			dH:12.069(5SF)	0	MCL	437

Reaction No. 54496

Zr+4_F-1(4) + F-1 = Zr+4_F-1(5)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1.51(0.06SD)	0	EOD	7033
3	25	1	NaClO4	lgK:1.9(0.05SD)	0	MQH	931
4	25	0	Inf. Dilution	dH:-4.0(5.SD)	0	MCL	437
5	25			dH:4.302(4SF)	0	MCL	437

Reaction No. 55129

SO4-2 + Zr+4 = Zr+4_SO4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:2.63(3SF)	5	MIX	13428
2	20	2	HClO4	lgK:2.56(3SF)	5	MSC	19258
3	20	4	HClO4	lgK:2.78(3SF)	3	MSE	13458
4	20	4	Unknown	lgK:3.68(3SF)	3	CRV	2556
5	23	2	HClO4	lgK:2.67(3SF)	3	MIX	3262
6	25	0	Inf. Dilution	lgK:3.8(0.05SD)	0	ENS	5458
7	25	0	Inf. Dilution	lgK:7.040(0.090MD)	3	CRV	20829
8	25	0	Inf. Dilution	lgK:3.61(3SF)	0	EOD	22958
9	25	0	Inf. Dilution	lgK:7.04(0.09SD)	3	EOD	31756
10	25	2	HClO4	lgK:3.74(3SF)	0	MDS	140
11	25	2	HClO4	lgK:2.66(3SF)	5	MSE	7998
12	25	2	HClO4	lgK:2.63(3SF)	5	MSE	13652
13	25	2	NaClO4	lgK:3.79(3SF)	0	MDS	8136
14	25	2	Unknown	lgK:3.67(3SF)	3	CRV	2556
15	25	4	HClO4	lgK:2.85(3SF)	5	MCE	22616
16	25	4	NaClO4	lgK:1.60(3SF)	0	MDS	4708
17	25			lgK:3.79(3SF)	4	CRV	32224

Reaction No. 55130

Zr+4 + 4<F-1> = Zr+4_F-1(4)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.2(3SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:18.7(0.05SD)	0	ENS	5458
3	25	0	Inf. Dilution	lgK:18.72(4SF)	0	CRV	32224

Reaction No. 55131

Zr+4 + 4<Cl-1> = Zr+4_Cl-1(4)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-37.5(0.05SD)	0	ENS	5458
3	25	0	Inf. Dilution	lgK:-37.53(4SF)	0	CRV	32224
4	25	0	Inf. Dilution	dH:-70.(2.SD)	3	MCL	5106

Reaction No. 55132

Zr+4 + 2<H2O> = ZrO2(s) + 4<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-1.9(0.05SD)	0	ENS	5458
3	25	0	Inf. Dilution	lgK:1.9(1.SD)	0	CRV	6478
Note (3): Sign positive for rxn as written in JPD							
4	25	0	Inf. Dilution	lgK:7.000(1.600MD)	0	CRV	20829
5	25	0	Inf. Dilution	lgK:1.9(2SF)	0	EOD	29487
6	25	0	Inf. Dilution	lgK:7.54(3SF)	4	CRV	32222
7	25	0	Inf. Dilution	lgK:-1.86(3SF)	0	CRV	32224
8	90	0	Inf. Dilution	lgK:1.56(3SF)	0	EOD	29487
9	25	0	Inf. Dilution	dH:-10.845(5SF)kJ	0	EOD	29487

Reaction No. 55133

Zr+4 + H2O = Zr+4_OH-1 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.0(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:0.3(0.05SD)	0	ENS	5458
3	25	0	Inf. Dilution	lgK:0.320(0.220MD)	2	CRV	20829
4	25	0	Inf. Dilution	lgK:0.3(1SF)	3	CRV	23624
5	25	0	Inf. Dilution	lgK:0.27(0.13SD)	3	CRV	32222
6	25	0	Inf. Dilution	lgK:0.3(1SF)	0	CRV	32224
7	25	0.7	Seawater	lgK:-0.24(2SF)	0	ENS	6496
8	25	4	ClO4-1 salt	lgK:-0.6(0.2SD)	0	MIS	5077
9	20	4	HClO4	lgR:0.3(0.3SD)	3	MSE	5077
10	25	0	Inf. Dilution	dH:9.8(2SF)	3	CRV	6478
11	25	0.7	Seawater	dH:9.8(2SF)	0	ENS	6496

Reaction No. 55136

4<Zr+4> + 8<OH-1> = Zr+4(4)_OH-1(8)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:118.5(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:106.0(0.1SD)	0	CSM	8
Note (2): Low wgt for inter-reaction consistency							
3	25	2	Unknown	lgK:103.4(0.05SD)	0	CSM	8

Reaction No. 55137

3<Zr+4> + 4<OH-1> = Zr+4(3)_OH-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:56.09(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:55.4(3SF)	0	CRV	2556
3	25	2	Unknown	lgK:50.5(0.05SD)	0	CSM	8

Reaction No. 55138

3<Zr+4> + 4<H2O> = Zr+4(3)_OH-1(4) + 4<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.6(0.05SD)	2	ENS	5458
2	25	0	Inf. Dilution	lgK:0.400(0.300MD)	3	CRV	20829
Note (2): Found not to be required by [18RKA_31367]							
3	25	0	Inf. Dilution	lgK:-0.6(1SF)	2	CRV	32224

Reaction No. 55139

Zr+4 + 5<H2O> = Zr+4_OH-1(5) + 5<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-19.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-16.0(0.05SD)	0	ENS	5458
3	25	0	Inf. Dilution	lgK:-17.97(0.15SD)	4	CRV	32222
4	25	0	Inf. Dilution	lgK:-16.0(3SF)	0	CRV	32224
5	25	0.7	Seawater	lgK:-17.07(4SF)	0	ENS	6496
6	25	0.7	Seawater	dH:32.6(3SF)	2	ENS	6496

Reaction No. 55140

4<Zr+4> + 8<H2O> = Zr+4(4)_OH-1(8) + 8<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.0(0.05SD)	2	ERS	5458
2	25	0	Inf. Dilution	lgK:6.520(0.650MD)	3	CRV	20829
Note (2): Found not to be required by [18RKA_31367]							

3	25	0	Inf. Dilution	lgK:6.0(2SF)	2	CRV	32224

Reaction No. 55351							
Zr+4 + OH-1 = Zr+4_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.8(3SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:14.3(3SF)	3	CRV	2556
3	25	0	Inf. Dilution	lgK:14.35(0.09SD)	3	EOD	7033
4	25	0	Inf. Dilution	lgK:14.2(3SF)	2	EOD	16981
5	25	0	Inf. Dilution	lgK:14.29(4SF)	3	EOD	29848
6	25	0	Inf. Dilution	lgK:14.32(0.22SD)	3	EOD	31367
7	25	0.1	HClO4	lgK:14.08(4SF)	5	MDS	7033
8	25	0.3	Unknown	lgK:14.12(4SF)	4	MDS	7033
9	25	0.5	Unknown	lgK:14.15(4SF)	3	MDS	7033
10	25	1	(H,Na)ClO4	lgK:14.6(3SF)	0	MSE	4444
11	25	1	(H,Na)ClO4	lgK:14.30(4SF)	2	MDS	5076
12	25	1	ClO4-1 salt	lgK:14.6(0.05SD)	0	MDS	4444
Note (12): Suspect other species							
13	25	1	ClO4-1 salt	lgK:14.32(4SF)	2	MGL	5076
14	25	1	Unknown	lgK:14.30(4SF)	2	MDS	7033
15	25	4	Unknown	lgK:13.9(3SF)	5	CRV	2556

Reaction No. 55352							
Zr+4 + 2<OH-1> = Zr+4_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:28.26(0.56SD)	0	EOD	7033
3	25	0	Inf. Dilution	lgK:27.1(3SF)	0	EOD	16981
4	25	0	Inf. Dilution	lgK:26.66(4SF)	0	EOD	29848
5	25	0	Inf. Dilution	lgK:25.7(3SF)	0	CRV	31367
Note (5): <= 25.7							
6	25	1	ClO4-1 salt	lgK:28.8(0.05SD)	0	MDS	4444
7	25	1	ClO4-1 salt	lgK:28.26(4SF)	0	MGL	5076

Reaction No. 55353							
Zr+4 + 3<OH-1> = Zr+4_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:41.24(0.24SD)	4	EOD	7033
2	25	0	Inf. Dilution	lgK:35.85(4SF)	0	EOD	29848
3	25	1	ClO4-1 salt	lgK:42.7(0.08SD)	4	MDS	4444
4	25	1	ClO4-1 salt	lgK:41.92(4SF)	4	MGL	5076

Reaction No. 55354

Zr+4 + 4<OH-1> = Zr+4_OH-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:53.81(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:54.38(0.57SD)	2	EOD	7033
3	25	0	Inf. Dilution	lgK:43.12(4SF)	0	EOD	29848
4	25	0	Inf. Dilution	lgK:45.89(4SF)	0	CRV	31367
Note (4): < 45.89							
5	25	1	ClO4-1 salt	lgK:56.5(0.1SD)	2	MDS	4444
6	25	1	ClO4-1 salt	lgK:55.29(4SF)	2	MGL	5076

Reaction No. 55355

Zr+4 + 3<NO3-1> = Zr+4_NO3-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	HNO3	lgR:-0.26(0.05SD)	5	MDS	5502

Reaction No. 55356

Zr+4 + 4<NO3-1> = Zr+4_NO3-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.9(2SF)	1	ELC	
2	25	4	HNO3	lgR:-0.82(0.05SD)	5	MDS	5502

Reaction No. 55357

Zr+4 + 4<Cl-1> = Zr+4_Cl-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	6.54	HCl	lgR:1.18(0.08SD)	5	MDS	5502

Reaction No. 55937

Ca(s) + Zr(s) + 1.5<O2(g)> = CaO.ZrO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:285.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:294.5(4SF)	0	ENS	6422

3	27	0	Inf. Dilution	lgK:292.6(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:215.7(4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:169.6(4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:138.9(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-401.8(0.05SD)	0	ENS	802
8	25	0	Inf. Dilution	dG:-401.8(4SF)	0	EFE	6422
9	25	0	Inf. Dilution	dH:-422.3(0.1SD)	2	ENS	802
10	25	0	Inf. Dilution	dH:-422.3(4SF)	2	MTD	6422
11	127	0	Inf. Dilution	dH:-422.1(4SF)	0	MTD	6422
12	227	0	Inf. Dilution	dH:-421.8(4SF)	0	MTD	6422
13	327	0	Inf. Dilution	dH:-421.4(4SF)	0	MTD	6422

Reaction No. 56354

Ba(s) + Zr(s) + 1.5<O2(g)> = BaZrO3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:287.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:296.9(4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:295.0(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:217.5(4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:171.1(4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:140.1(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-405.0(0.05SD)	0	ENS	802
8	25	0	Inf. Dilution	dG:-405.0(4SF)	0	EFE	6422
9	25	0	Inf. Dilution	dH:-425.3(0.1SD)	0	ENS	802
10	25	0	Inf. Dilution	dH:-425.3(4SF)	0	MTD	6422
11	127	0	Inf. Dilution	dH:-425.1(4SF)	0	MTD	6422
12	227	0	Inf. Dilution	dH:-425.0(4SF)	0	MTD	6422
13	327	0	Inf. Dilution	dH:-425.0(4SF)	0	MTD	6422

Reaction No. 56395

Sr(s) + Zr(s) + 1.5<O2(g)> = SrZrO3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:294.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:292.7(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:215.8(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:169.7(4SF)	1	ENS	6422

5	327	0	Inf. Dilution	lgK:139.0(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-402.2(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-402.2(0.05SD)	5	ENS	802
8	25	0	Inf. Dilution	dG:-401.9(4SF)	4	EFE	6422
9	25	0	Inf. Dilution	dH:-422.4(0.1SD)	4	ENS	802
10	25	0	Inf. Dilution	dH:-422.4(0.1SD)	5	ENS	802
11	25	0	Inf. Dilution	dH:-422.4(4SF)	4	MTD	6422
12	127	0	Inf. Dilution	dH:-422.1(4SF)	1	MTD	6422
13	227	0	Inf. Dilution	dH:-421.7(4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-421.2(4SF)	0	MTD	6422

Reaction No. 56396							
Zr(s) + 2<F2(g)> = Zr+4_F-1(4)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:317.1(4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:315.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:231.9(4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:182.0(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:148.9(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-432.6(0.05SD)	0	ENS	802
7	25	0	Inf. Dilution	dG:-432.6(0.05SD)	0	ENS	802
8	25	0	Inf. Dilution	dG:-432.6(4SF)	0	EFE	6422
9	25	0	Inf. Dilution	dH:-456.8(0.1SD)	0	ENS	802
10	25	0	Inf. Dilution	dH:-456.8(0.1SD)	0	ENS	802
11	25	0	Inf. Dilution	dH:-456.8(4SF)	0	MTD	6422
12	127	0	Inf. Dilution	dH:-456.3(4SF)	0	MTD	6422
13	227	0	Inf. Dilution	dH:-455.8(4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-455.2(4SF)	0	MTD	6422

Reaction No. 56397							
2<O2(g)> + Si(s) + Zr(s) = ZrSiO4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:334.5(4SF)	2	ENS	6422
2	25	0	Inf. Dilution	lgK:336.3(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:332.3(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:334.1(4SF)	0	ENS	6494

5	127	0	Inf. Dilution	lgK:244.2(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:245.6(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:191.4(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:192.5(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:156.2(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:157.1(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-458.7(0.05SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-458.7(0.05SD)	5	ENS	802
13	25	0	Inf. Dilution	dG:-456.3(4SF)	0	EFE	6422
14	25	0	Inf. Dilution	dG:-458.8(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dG:-1919.7(3.1MD)kJ	6	CRV	20829
16	25	0	Inf. Dilution	dH:-486.0(0.1SD)	4	ENS	802
17	25	0	Inf. Dilution	dH:-486.0(0.1SD)	5	ENS	802
18	25	0	Inf. Dilution	dH:-483.7(4SF)	2	MTD	6422
19	25	0	Inf. Dilution	dH:-486.2(4SF)	4	MTD	6494
20	25	0	Inf. Dilution	dH:-2034.2(3.1MD)kJ	6	CRV	20829
21	127	0	Inf. Dilution	dH:-483.7(4SF)	1	MTD	6422
22	127	0	Inf. Dilution	dH:-486.1(4SF)	1	MTD	6494
23	227	0	Inf. Dilution	dH:-483.4(4SF)	0	MTD	6422
24	227	0	Inf. Dilution	dH:-485.9(4SF)	0	MTD	6494
25	327	0	Inf. Dilution	dH:-483.1(4SF)	0	MTD	6422
26	327	0	Inf. Dilution	dH:-485.5(4SF)	0	MTD	6494
27	25	0	Inf. Dilution	Cp:98.200(0.6MD)J/K	5	CRV	20829

Reaction No. 57698

2<S(s)> + Zr(s) = Zr+4_S-2(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:99.86(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:99.24(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:74.08(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:58.84(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:48.62(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-136.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-565.750(12.0MD)kJ	6	CRV	20829
8	25	0	Inf. Dilution	dH:-138.0(4SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-573.200(12.0MD)kJ	6	CRV	20829

10	127	0	Inf. Dilution	dH:-139.1(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-139.9(4SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-140.6(4SF)	0	MTD	6422
13	25	0	Inf. Dilution	Cp:56.400(2.0MD)J/K	6	CRV	20829

Reaction No. 58378

Zr+4_NO3-1 + NO3-1 = Zr+4_NO3-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:30	2	HClO4	lgK:-0.22(2SF)	4	MIX	22619
2	18:30	4	HClO4	lgK:-0.81(2SF)	4	MIX	22619
3	20	2	HClO4	lgK:-0.45(2SF)	4	MIX	13428
4	20	4	HClO4	lgK:-0.23(2SF)	0	MDS	140
5	23	2	HClO4	lgK:-0.30(2SF)	3	MIX	3262
6	23	4	HClO4	lgK:-0.80(2SF)	3	MIX	3262
7	25	0	Inf. Dilution	lgK:2.1(2SF)	1	EDH	
8	25	2	HClO4	lgK:-0.46(2SF)	5	MIX	13428
9	25	4	HClO4	lgK:0.11(2SF)	0	MIX	5502

Reaction No. 58379

Zr+4_NO3-1(2) + NO3-1 = Zr+4_NO3-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgK:-0.37(2SF)	3	MDS	140
2	25	4	HClO4	lgK:-0.26(2SF)	5	MIX	5502

Reaction No. 58380

Zr+4_NO3-1(3) + NO3-1 = Zr+4_NO3-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgK:-0.56(2SF)	3	MDS	140
2	25	4	HClO4	lgK:-0.82(2SF)	5	MIX	5502

Reaction No. 58381

5<NO3-1> + Zr+4 = Zr+4_NO3-1(5)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgK:-1.5(2SF)	3	MSC	140
2	25	0	Inf. Dilution	lgK:-0.803(3SF)	2	EAE	

Reaction No. 58382

6<NO3-1> + Zr+4 = Zr+4_NO3-1(6)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgK:-1.7(2SF)	3	MSC	140
2	25	0	Inf. Dilution	lgK:-1.07(3SF)	2	EAE	

Reaction No. 58383

2<NO3-1> + ZrO+2 = ZrO+2_NO3-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2.5	HClO4	lgK:1.91(3SF)	5	MDS	140

Reaction No. 58384

4<NO3-1> + ZrO+2 = ZrO+2_NO3-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2.5	HClO4	lgK:1.63(3SF)	5	MDS	140

Reaction No. 58511

Zr+4_SO4-2 + SO4-2 = Zr+4_SO4-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:1.88(3SF)	3	MIX	13428
2	20	2	HClO4	lgK:0.78(2SF)	0	MSC	19258
3	20	4	HClO4	lgK:1.95(3SF)	5	MSE	13458
4	23	2	HClO4	lgK:3.54(3SF)	0	MIX	3262
5	25	0	Inf. Dilution	lgK:5.1(2SF)	1	EES	
6	25	2	HClO4	lgK:2.80(3SF)	0	MDS	140
7	25	2	HClO4	lgK:1.72(3SF)	5	MSE	7998
8	25	2	HClO4	lgK:1.73(3SF)	5	MSE	13652
9	25	2	NaClO4	lgK:2.85(3SF)	0	MDS	8136
10	25	4	HClO4	lgK:1.85(3SF)	5	MCE	22616
11	25	4	NaClO4	lgK:1.12(3SF)	0	MDS	4708

Reaction No. 58512

Zr+4_SO4-2(2) + SO4-2 = Zr+4_SO4-2(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:1.00(3SF)	3	MIX	13428
2	20	2	HClO4	lgK:2.27(3SF)	0	MSC	19258
3	20	4	HClO4	lgK:0.30(2SF)	5	MSE	13458
4	23	2	HClO4	lgK:6.59(3SF)	0	MIX	3262

5	25	0	Inf. Dilution	lgK:1.7(2SF)	1	EDH	
6	25	2	HClO4	lgK:1.1(2SF)	3	MDS	140
7	25	2	HClO4	lgK:0.0(0.3MD)	0	MSE	7998
8	25	2	HClO4	lgK:-0.08(1SF)	0	MSE	13652

Reaction No. 58574							
Zr+4_Cl-1 + Cl-1 = Zr+4_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:30	2	HClO4	lgK:-0.40(2SF)	6	CRV	20829
2	18:30	2	HClO4	lgK:-0.90(2SF)	4	MIX	22619
3	18:30	4	HClO4	lgK:-0.69(2SF)	6	CRV	20829
4	18:30	4	HClO4	lgK:-0.72(2SF)	4	MIX	22619
5	20	2	HClO4	lgK:-0.61(2SF)	4	MIX	13428
6	20	6.54	H(ClO4)	lgK:0.40(2SF)	3	MDS	140
7	23	2	HClO4	lgK:-0.90(2SF)	3	MIX	3262
8	23	4	HClO4	lgK:-0.74(2SF)	3	MIX	3262
9	25	0	Inf. Dilution	lgK:-0.65(2SF)	2	EOD	7033
10	25	6.5	HClO4	lgK:1.32(3SF)	3	MIX	5502

Reaction No. 58575							
Zr+4_Cl-1(2) + Cl-1 = Zr+4_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:30	2	HClO4	lgK:-0.23(2SF)	4	MIX	22619
2	18:30	4	HClO4	lgK:-0.58(2SF)	6	CRV	20829
3	18:30	4	HClO4	lgK:-0.62(2SF)	4	MIX	22619
4	20	2	HClO4	lgK:-0.46(2SF)	4	MIX	13428
5	20	6.54	H(ClO4)	lgK:0.19(2SF)	3	MDS	140
6	23	2	HClO4	lgK:-0.23(2SF)	3	MIX	3262
7	23	4	HClO4	lgK:-0.62(2SF)	3	MIX	3262
8	25	0	Inf. Dilution	lgK:-1.08(0.08SD)	2	EOD	7033

Reaction No. 58576							
Zr+4_Cl-1(3) + Cl-1 = Zr+4_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:30	2	HClO4	lgK:0.06(1SF)	4	MIX	22619
2	18:30	4	HClO4	lgK:-0.30(2SF)	6	CRV	20829
3	20	6.54	H(ClO4)	lgK:-0.33(2SF)	3	MDS	140

4	23	2	HClO4	lgK:0.06(1SF)	3	MIX	3262
5	25	0	Inf. Dilution	lgK:-1.10(3SF)	2	EOD	7033

Reaction No. 58713

ZrO+2_H+1(-1)_Arg-1_OH-1 + H+1 = ZrO+2_H+1(-1)_Arg-1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:3.9(2SF)	0	MGL	140

Reaction No. 58714

ZrO+2_H+1(-1)_Arg-1_OH-1(2) + H+1 = ZrO+2_H+1(-1)_Arg-1_OH-1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:8.5(2SF)	0	MGL	140

Reaction No. 59997

Zr+4 + H+1_Ala-1 = Zr+4_H+1_Ala-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	Unknown	lgK:4.1(0.1SD)	3	MSP	2392
2	25	0	Inf. Dilution	lgK:4.1(3SF)	2	EAE	

Reaction No. 60080

Zr+4 + H+1_bAla-1 = Zr+4_H+1_bAla-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	Unknown	lgK:5.3(0.1SD)	3	MSP	2392
2	25	0	Inf. Dilution	lgK:5.4(2SF)	1	EAE	

Reaction No. 60177

Zr+4_EDTA-4 + H+1 = Zr+4_H+1_EDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:6.1(2SF)	4	MRX	142
2	25	0	Inf. Dilution	lgK:6.1(3SF)	2	EAE	

Reaction No. 60502

Zr+4 + 5<OH-1> = Zr+4_OH-1(5)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:52.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:54.0(3SF)	0	CRV	2556
3	25	0	Inf. Dilution	lgK:46.52(4SF)	0	EOD	29848

4	25	0	Inf. Dilution	lgK:50.35(0.23SD)	0	CRV	31367

Reaction No. 60503							
ZrO2(s) + 2<H2O> = Zr+4 + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-63.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-54.1(3SF)	0	CRV	2556
3	25	1	Unknown	lgK:-52.0(3SF)	0	CRV	2556

Reaction No. 60601							
O2(g) + Zr(s) = ZrO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:182.2(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:182.7(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:181.0(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:181.5(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:133.2(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:133.6(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:104.6(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:104.9(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:85.49(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:85.76(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-248.5(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-249.3(4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dG:-1042.7(1.3MD)kJ	6	CRV	20829
14	25	0	Inf. Dilution	dG:-70.510(5SF)	0	CRV	31291
15	25	0	Inf. Dilution	dH:-262.3(4SF)	2	MTD	6422
16	25	0	Inf. Dilution	dH:-263.0(4SF)	2	MTD	6494
17	25	0	Inf. Dilution	dH:-1100.60(1.3MD)kJ	6	CRV	20829
18	127	0	Inf. Dilution	dH:-262.2(4SF)	1	MTD	6422
19	127	0	Inf. Dilution	dH:-262.9(4SF)	1	MTD	6494
20	227	0	Inf. Dilution	dH:-262.0(4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-262.7(4SF)	0	MTD	6494
22	327	0	Inf. Dilution	dH:-261.7(4SF)	0	MTD	6422
23	327	0	Inf. Dilution	dH:-262.5(4SF)	0	MTD	6494
24	25	0	Inf. Dilution	Cp:55.960(0.8MD)J/K	6	CRV	20829

Reaction No. 60630							
2<SO4-2> + Zr+4 = Zr+4_SO4-2(2)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	20	4	Unknown	lgK:6.4(2SF)	0	CRV	2556
2	25	0	Inf. Dilution	lgK:11.540(0.210MD)	3	CRV	20829
3	25	0	Inf. Dilution	lgK:11.54(4SF)	3	EOD	31756
4	25	2	NaClO4	lgK:6.54(3SF)	0	MDS	7998
5	25	2	NaClO4	lgK:6.64(3SF)	0	MDS	8136
6	25	2	Unknown	lgK:6.40(3SF)	0	CRV	2556
7	25	4	NaClO4	lgK:2.72(3SF)	0	MDS	4708

Reaction No. 60631							
3<SO4-2> + Zr+4 = Zr+4_SO4-2(3)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	20	4	Unknown	lgK:7.5(2SF)	0	CRV	2556
2	25	0	Inf. Dilution	lgK:14.3(3SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:14.300(0.500MD)	0	CRV	20829
4	25	2	Unknown	lgK:7.4(2SF)	0	CRV	2556

Reaction No. 60676							
2<F-1> + Zr+4 = Zr+4_F-1(2)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:11.9(3SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:18.550(0.310MD)	3	CRV	20829
3	25	2	Unknown	lgK:16.4(3SF)	0	CRV	2556
4	25	4	Unknown	lgK:17.23(4SF)	0	CRV	2556
5	25	0	Inf. Dilution	dH:-16.800(1.00MD)kJ	3	CRV	20829
6	25	4	Unknown	dH:-2.5(2SF)	0	CRV	2556

Reaction No. 60677							
3<F-1> + Zr+4 = Zr+4_F-1(3)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:18.1(3SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:24.720(0.380MD)	3	CRV	20829
3	25	2	Unknown	lgK:22.4(3SF)	0	CRV	2556
4	25	4	Unknown	lgK:23.8(3SF)	0	CRV	2556
5	25	0	Inf. Dilution	dH:-11.200(1.70MD)kJ	3	CRV	20829

6	25	4	Unknown	dH:-2.3(2SF)	0	CRV	2556

Reaction No. 60678							
Zr+4 + 4<F-1> = Zr+4_F-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.0(3SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:30.110(0.400MD)	3	CRV	20829
3	25	4	Unknown	lgK:29.5(3SF)	0	CRV	2556
4	25	0	Inf. Dilution	dH:-22.000(2.700MD)kJ	3	CRV	20829
5	25	4	Unknown	dH:-4.7(2SF)	0	CRV	2556

Reaction No. 60679							
5<F-1> + Zr+4 = Zr+4_F-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	Unknown	lgK:34.6(3SF)	0	CRV	2556
2	25	0	Inf. Dilution	lgK:27.3(3SF)	0	ELC	
3	25	0	Inf. Dilution	lgK:34.600(0.420MD)	3	CRV	20829

Reaction No. 60886							
ZrO+2 + H+1_24DHB-3 = ZrO+2_H+1_24DHB-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	30	0.1	KNO3	lgK:16.55(4SF)	6	MGL	5987

Reaction No. 61466							
Zr+4 + H+1_Gluconic-1 = Zr+4_Gluconic-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	NaClO4	lgK:1.73(3SF)	4	MIX	2261

Reaction No. 61487							
Zr+4_NTA-3 + NTA-3 = Zr+4_NTA-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:7.8(0.2SD)	0	MSP	1118
2	25	0	Inf. Dilution	lgK:7.8(2SF)	1	ESC	

Reaction No. 66213							
2<I2(s)> + Zr(s) = Zr+4_I-1(4)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:85.05(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:84.52(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:63.09(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:48.60(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:38.06(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-116.0(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-480.860(3.7MD)kJ	6	CRV	20829
8	25	0	Inf. Dilution	dG:-115.30(5SF)	3	CRV	31291
9	25	0	Inf. Dilution	dH:-116.8(4SF)	4	MTD	6422
10	25	0	Inf. Dilution	dH:-488.900(3.7MD)kJ	6	CRV	20829
11	25	0	Inf. Dilution	dH:-116.80(5SF)	3	CRV	31291
12	127	0	Inf. Dilution	dH:-124.7(4SF)	1	MTD	6422
13	227	0	Inf. Dilution	dH:-145.2(4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-144.4(4SF)	0	MTD	6422
15	25	0	Inf. Dilution	Cp:125.100(0.5MD)J/K	6	CRV	20829

Reaction No. 66214

1.5<I2(s)> + Zr(s) = Zr+3_I-1(3)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:69.19(4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:68.76(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:51.34(4SF)	1	ENS	6422
4	227	0	Inf. Dilution	lgK:39.68(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:31.23(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-94.40(4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dG:-385.580(15.0MD)kJ	0	CRV	20829
8	25	0	Inf. Dilution	dG:-92.986(5SF)	2	CRV	31291
9	25	0	Inf. Dilution	dH:-95.0(3SF)	0	MTD	6422
10	25	0	Inf. Dilution	dH:-391.600(15.0MD)kJ	0	CRV	20829
11	25	0	Inf. Dilution	dH:-93.600(5SF)	1	CRV	31291
12	127	0	Inf. Dilution	dH:-100.9(4SF)	0	MTD	6422
13	227	0	Inf. Dilution	dH:-116.2(4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-115.7(4SF)	0	MTD	6422
15	25	0	Inf. Dilution	Cp:100.300(0.2MD)J/K	0	CRV	20829

Reaction No. 66215

I2(s) + Zr(s) = Zr+2_I-1(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:45.20(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:44.92(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:33.58(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:26.01(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:20.54(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-61.66(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-273.800(10.0MD)kJ	0	CRV	20829
8	25	0	Inf. Dilution	dG:-66.049(5SF)	0	CRV	31291
9	25	0	Inf. Dilution	dH:-62.0(3SF)	1	MTD	6422
10	25	0	Inf. Dilution	dH:-277.800(10.0MD)kJ	3	CRV	20829
11	25	0	Inf. Dilution	dH:-66.400(5SF)	1	CRV	31291
12	127	0	Inf. Dilution	dH:-65.5(3SF)	3	MTD	6422
13	227	0	Inf. Dilution	dH:-75.4(3SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-74.6(3SF)	0	MTD	6422
15	25	0	Inf. Dilution	Cp:75.500(0.2MD)J/K	6	CRV	20829

Reaction No. 66216

Zr(s) + 1.5<F2(g)> = Zr+3_F-1(3)_(s)

Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:232.2(4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:230.7(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:169.7(4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:133.2(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:108.9(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-316.8(4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dH:-335.0(4SF)	0	MTD	6422
8	127	0	Inf. Dilution	dH:-334.6(4SF)	0	MTD	6422
9	227	0	Inf. Dilution	dH:-334.2(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-333.7(4SF)	0	MTD	6422

Reaction No. 66217

Zr(s) + F2(g) = Zr+2_F-1(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:159.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:158.9(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:117.0(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:91.90(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:75.20(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-218.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-230.0(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-229.8(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-229.5(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-229.2(4SF)	0	MTD	6422

Reaction No. 66218

2<Cl2(g)> + Zr(s) = Zr+4_Cl-1(4)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:155.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:154.7(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:112.1(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:86.66(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:69.73(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-889.9(4SF)kJ	3	CRV	6330
7	25	0	Inf. Dilution	dG:-212.5(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dG:-890.150(1.2MD)kJ	4	CRV	20829
9	25	0	Inf. Dilution	dH:-980.5(4SF)kJ	3	CRV	6330
10	25	0	Inf. Dilution	dH:-234.2(4SF)	4	MTD	6422
11	25	0	Inf. Dilution	dH:-980.800(0.9MD)kJ	4	CRV	20829
12	127	0	Inf. Dilution	dH:-233.5(4SF)	4	MTD	6422
13	227	0	Inf. Dilution	dH:-232.8(4SF)	3	MTD	6422
14	327	0	Inf. Dilution	dH:-232.1(4SF)	0	MTD	6422
15	25	0	Inf. Dilution	Cp:117.500(1.6MD)J/K	6	CRV	20829

Reaction No. 66219

Zr(s) + 1.5<Cl2(g)> = Zr+3_Cl-1(3)_(s)

Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:113.2(4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:112.4(4SF)	0	ENS	6422

3	127	0	Inf. Dilution	lgK:81.40(4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:62.83(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:50.49(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-154.5(4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dG:-689.740(3.0MD)kJ	0	CRV	20829
8	25	0	Inf. Dilution	dH:-170.7(4SF)	0	MTD	6422
9	25	0	Inf. Dilution	dH:-760.100(3.0MD)kJ	0	CRV	20829
10	127	0	Inf. Dilution	dH:-170.2(4SF)	0	MTD	6422
11	227	0	Inf. Dilution	dH:-169.7(4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-169.1(4SF)	0	MTD	6422
13	25	0	Inf. Dilution	Cp:92.950(0.6MD)J/K	0	CRV	20829

Reaction No. 66220

Cl2(g) + Zr(s) = Zr+2_Cl-1(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:67.56(4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:67.10(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:48.37(4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:37.17(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:29.73(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-92.17(4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dG:-496.150(13.0MD)kJ	6	CRV	20829
8	25	0	Inf. Dilution	dH:-502.0(4SF)kJ	0	CRV	6330
9	25	0	Inf. Dilution	dH:-103.0(4SF)	0	MTD	6422
10	25	0	Inf. Dilution	dH:-542.800(13.0MD)kJ	6	CRV	20829
11	127	0	Inf. Dilution	dH:-102.7(4SF)	0	MTD	6422
12	227	0	Inf. Dilution	dH:-102.3(4SF)	0	MTD	6422
13	327	0	Inf. Dilution	dH:-101.9(4SF)	0	MTD	6422
14	25	0	Inf. Dilution	Cp:68.800(1.6MD)J/K	6	CRV	20829

Reaction No. 66221

Zr(s) + C(s) = ZrC(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.8(3SF)	1	EDH	
2	25	0	Inf. Dilution	lgK:33.86(4SF)	4	ENS	6422
3	27	0	Inf. Dilution	lgK:33.65(4SF)	0	ENS	6422

4	127	0	Inf. Dilution	lgK:25.10(4SF)	4	ENS	6422
5	227	0	Inf. Dilution	lgK:19.98(4SF)	4	ENS	6422
6	327	0	Inf. Dilution	lgK:16.58(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-46.20(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dG:-203.480(2.6MD)kJ	0	CRV	20829
9	25	0	Inf. Dilution	dH:-47.0(3SF)	0	MTD	6422
10	25	0	Inf. Dilution	dH:-207.000(2.5MD)kJ	6	CRV	20829
11	127	0	Inf. Dilution	dH:-46.9(3SF)	4	MTD	6422
12	227	0	Inf. Dilution	dH:-46.7(3SF)	4	MTD	6422
13	327	0	Inf. Dilution	dH:-46.6(3SF)	0	MTD	6422
14	25	0	Inf. Dilution	Cp:37.280(0.7MD)J/K	6	CRV	20829

Reaction No. 66222

2<Br2(l)> + Zr(s) = Zr+4_Br-1(4)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:127.1(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:126.2(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:91.53(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:70.17(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:55.97(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-173.3(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-760.7(4SF)kJ	3	CRV	6330
8	25	0	Inf. Dilution	dH:-181.8(4SF)	3	MTD	6422
9	25	0	Inf. Dilution	dH:-759.500(3.3MD)kJ	5	CRV	20829
10	127	0	Inf. Dilution	dH:-195.9(4SF)	0	MTD	6422
11	227	0	Inf. Dilution	dH:-195.1(4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-194.4(4SF)	0	MTD	6422

Reaction No. 66223

Zr(s) + 1.5<Br2(l)> = Zr+3_Br-1(3)_(s)

Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:106.4(4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:105.8(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:76.88(4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:59.14(4SF)	0	ENS	6422

5	327	0	Inf. Dilution	lgK:47.35(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-145.2(4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dH:-152.0(4SF)	0	MTD	6422
8	25	0	Inf. Dilution	dH:-599.800(13.0MD)kJ	0	CRV	20829
9	127	0	Inf. Dilution	dH:-162.6(4SF)	0	MTD	6422
10	227	0	Inf. Dilution	dH:-162.1(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-161.5(4SF)	0	MTD	6422

Reaction No. 66224

Zr(s) + Br2(l) = Zr+2_Br-1(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:66.96(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:66.52(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:48.13(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:36.86(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:29.38(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-91.35(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-96.7(3SF)	4	MTD	6422
8	25	0	Inf. Dilution	dH:-399.200(13.0MD)kJ	0	CRV	20829
9	127	0	Inf. Dilution	dH:-103.5(4SF)	1	MTD	6422
10	227	0	Inf. Dilution	dH:-102.9(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-102.3(4SF)	0	MTD	6422

Reaction No. 66264

2<Y(s)> + 2<Zr(s)> + 3.5<O2(g)> = Y2Zr2O7(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:686.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:681.9(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:502.6(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:395.0(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:323.3(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-936.4(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-985.2(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-984.7(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-983.9(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-983.0(4SF)	0	MTD	6422

Reaction No. 66418

2<Sm(s)> + 2<Zr(s)> + 3.5<O2(g)> = Sm2Zr2O7(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:677.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:673.2(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:496.0(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:389.8(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:319.0(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-924.3(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-973.0(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-972.4(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-971.7(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-971.0(4SF)	0	MTD	6422

Reaction No. 66842

ZrO+2_H+1(4)_PO4-3(2)_(s) + H2O = Zr+4 + 2<OH-1> + 2<H+1(2)_PO4-3>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:-17.64(4SF)	4	ENS	773
2	25	0	Inf. Dilution	lgK:-17.6(3SF)	2	EAE	

Reaction No. 66959

2<Li(s)> + Zr(s) + 1.5<O2(g)> = Li2ZrO3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:292.0(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:290.1(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:213.5(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:167.5(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:136.7(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-398.4(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-420.7(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-420.6(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-422.0(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-421.9(4SF)	0	MTD	6422

Reaction No. 67090

2<Nd(s)> + 3.5<O2(g)> + 2<Zr(s)> = Nd2O3.2ZrO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:673.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:669.2(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:493.1(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:387.5(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:317.2(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-918.9(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-967.2(4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-966.6(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-965.7(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-964.8(4SF)	0	MTD	6422

Reaction No. 67451

Zr+4 + 2<H2O> = Zr+4_OH-1(2) + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:0.980(1.060MD)	0	CRV	20829
3	25	0	Inf. Dilution	lgK:2.7(2SF)	0	CRV	32222
Note (3): < 2.7 (seems odd)							
4	25	0	Inf. Dilution	lgK:-1.7(2SF)	0	CRV	32224
5	25	0.7	Seawater	lgK:-2.79(3SF)	0	ENS	6496
Note (5): p. 984							
6	25	0.7	Seawater	dH:18.1(3SF)	0	ENS	6496
Note (6): p .984							

Reaction No. 67452

Zr+4 + 3<H2O> = Zr+4_OH-1(3) + 3<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.8(1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-4.5(2SF)	0	CRV	32222
Note (2): < -4.5							
3	25	0	Inf. Dilution	lgK:-5.1(2SF)	0	CRV	32224
4	25	0.7	Seawater	lgK:-6.48(3SF)	0	ENS	6496
5	25	0.7	Seawater	dH:24.7(3SF)	0	ENS	6496

Reaction No. 67453

Zr+4 + 4<H2O> = Zr+4_OH-1(4) + 4<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:-1.7(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-2.190(1.700MD)	0	CRV	20829
3	25	0	Inf. Dilution	lgK:-5.13(0.36SD)	0	EOD	32222
4	25	0	Inf. Dilution	lgK:-9.7(2SF)	0	CRV	32224
5	25	0.7	Seawater	lgK:-11.08(4SF)	0	ENS	6496
6	25	0.7	Seawater	dH:29.3(3SF)	0	ENS	6496

Reaction No. 70917

ZrO2(s) + 4<e-1> + 4<H+1> = Zr(s) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-99.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.553(4SF)	0	CRV	6330
3	25	0	Inf. Dilution	E:-1.473(4SF)	0	CRV	7761
4	25	0	Inf. Dilution	E:-1.553(4SF)	0	CRV	22519
5	25	0	Inf. Dilution	E:-1.45(3SF)	0	CRV	22551
6	25	0	Inf. Dilution	dH:528.9(4SF)kJ	0	CRV	7761

Reaction No. 71654

Zr+4_OH-1 + H+1 + 4<e-1> = Zr(s) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-93.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.45(3SF)	0	CRV	7761

Reaction No. 71655

Zr+4(4)_OH-1(8) + 8<H+1> + 16<e-1> = 4<Zr(s)> + 8<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-378.7(4SF)	0	ELC	
Note (1): from DOCMN using only SSS							
2	25	0	Inf. Dilution	lgK:-374.6(4SF)	1	ELC	
3	25	0	Inf. Dilution	E:-1.47(3SF)	0	CRV	7761

Reaction No. 71656

Zr+4_OH-1(5) + 4<e-1> = Zr(s) + 5<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-144.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.22(3SF)	0	CRV	7761

Reaction No. 71657

ZrO2(am.,s) + 2<H2O> + 4<e-1> = Zr(s) + 4<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-148.2(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.28(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:743.0(4SF)kJ	2	CRV	7761

Reaction No. 71658

ZrO2(s) + 2<H2O> + 4<e-1> = Zr(s) + 4<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-155.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.301(4SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:752.3(4SF)kJ	2	CRV	7761

Reaction No. 73371

Zr+4_OH-1 + OH-1 = Zr+4_OH-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.9(3SF)	2	EDH	
2	25	1	(H,Na)ClO4	lgK:14.2(3SF)	3	MSE	4444
3	25	1	ClO4-1 salt	lgK:13.94(4SF)	4	MGL	5076

Reaction No. 73372

Zr+4_OH-1(2) + OH-1 = Zr+4_OH-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.3(3SF)	1	ELC	
2	25	1	(H,Na)ClO4	lgK:13.9(3SF)	0	MSE	4444
3	25	1	ClO4-1 salt	lgK:13.65(4SF)	0	MGL	5076

Reaction No. 73373

Zr+4_OH-1(3) + OH-1 = Zr+4_OH-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.6(3SF)	1	ELC	
2	25	1	(H,Na)ClO4	lgK:13.8(3SF)	0	MSE	4444
3	25	1	ClO4-1 salt	lgK:13.36(4SF)	0	MGL	5076

Reaction No. 73876

ZrO(OH)2(s) + H2O + 4<e-1> = Zr(s) + 4<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.36(3SF)	5	CRV	22519

Reaction No. 74031							
Zr+4_OH-1(4)_(s) = Zr+4 + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-59.24(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-52.0(3SF)	0	EOD	16981
3	25	0	Inf. Dilution	lgK:-56.(2SF)	0	CRV	22551
4	25	0	Inf. Dilution	lgK:-56.94(4SF)	0	EOD	29848

Reaction No. 74536							
Zr(s) = Zr(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:566.880(1.4MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:609.800(1.4MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:24.775(0.01MD)J/K	6	CRV	20829

Reaction No. 74537							
0.5<H2(g)> + 0.5<O2(g)> + Zr(s) - 3<e-1> = Zr+4_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:134.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-767.480(9.312MD)kJ	0	CRV	20829

Reaction No. 74538							
F2(g) + Zr(s) = ZrF2(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-492.770(1.436MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-478.300(0.800MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:49.400(2.000MD)J/K	6	CRV	20829

Reaction No. 74539							
0.5<O2(g)> + Zr(s) = ZrO(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:30.576(27.1MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:56.200(27.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:30.800(1.0MD)J/K	6	CRV	20829

Reaction No. 74540

O2(g) + Zr(s) = ZrO2(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-298.390(47.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-289.600(47.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:46.100(1.0MD)J/K	6	CRV	20829

Reaction No. 74541

0.5<H2(g)> + Zr(s) = ZrH(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-64.686(0.8MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-84.400(0.8MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:30.200(0.1MD)J/K	6	CRV	20829

Reaction No. 74542

H2(g) + Zr(s) = ZrH2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-124.280(1.7MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-164.400(1.7MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:30.930(0.06MD)J/K	6	CRV	20829

Reaction No. 74543

H2(g) + O2(g) + Zr(s) - 2<e-1> = Zr+4_OH-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:177.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1008.4(11.0MD)kJ	0	CRV	20829

Reaction No. 74544

2<H2(g)> + 2<O2(g)> + Zr(s) = Zr+4_OH-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:256.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1464.6(13.3MD)kJ	0	CRV	20829

Reaction No. 74545

3<H2(g)> + 3<O2(g)> + Zr(s) + 2<e-1> = Zr+4_OH-1(6)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:308.2(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1785.8(10.1MD)kJ	0	CRV	20829

Reaction No. 74546							
2<H2(g)> + 2<O2(g)> + 3<Zr(s)> - 8<e-1> = Zr+4(3)_OH-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:442.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2536.4(27.7MD)kJ	0	CRV	20829
3	25	0	Inf. Dilution	dH:-2970.8(10.0MD)kJ	2	CRV	20829
4	25	0	Inf. Dilution	Cp:-1190.9(98.8MD)J/K	2	CRV	20829

Reaction No. 74547							
4.5<H2(g)> + 4.5<O2(g)> + 3<Zr(s)> - 3<e-1> = Zr+4(3)_OH-1(9)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:665.3(4SF)	0	ELC	
Note (1): from DOCMN using only SSS							
2	25	0	Inf. Dilution	lgK:662.3(4SF)	1	ELC	
3	25	0	Inf. Dilution	dG:-3789.4(27.7MD)kJ	0	CRV	20829

Reaction No. 74548							
4<H2(g)> + 4<O2(g)> + 4<Zr(s)> - 8<e-1> = Zr+4(4)_OH-1(8)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:711.1(4SF)	0	ELC	
Note (1): from DOCMN using only SSS							
2	25	0	Inf. Dilution	lgK:707.0(4SF)	1	ELC	
3	25	0	Inf. Dilution	dG:-4048.4(37.1MD)kJ	0	CRV	20829

Reaction No. 74549							
7.5<H2(g)> + 7.5<O2(g)> + 4<Zr(s)> - e-1 = Zr+4(4)_OH-1(15)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1008.0(5SF)	0	ELC	
Note (1): from DOCMN using only SSS							
2	25	0	Inf. Dilution	lgK:1004.(4SF)	1	ELC	
3	25	0	Inf. Dilution	dG:-5743.0(36.9MD)kJ	0	CRV	20829

Reaction No. 74550
8<H2(g)> + 8<O2(g)> + 4<Zr(s)> = Zr+4(4)_OH-1(16)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1045.0(5SF)	0	ELC	
Note (1): from DOCMN using only SSS							
2	25	0	Inf. Dilution	lgK:1041.(4SF)	1	ELC	
3	25	0	Inf. Dilution	dG:-5956.2(37.2MD)kJ	0	CRV	20829
4	25	0	Inf. Dilution	dH:-6706.20(7.2MD)kJ	0	CRV	20829

Reaction No. 74551

0.5<F2(g)> + Zr(s) = ZrF(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:41.353(1.2MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:72.100(1.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:32.800(1.0MD)J/K	6	CRV	20829

Reaction No. 74552

0.5<F2(g)> + Zr(s) - 3<e-1> = Zr+4_F-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:148.3(4SF)	0	ELC	
2	25	0	Inf. Dilution	dG:-867.800(9.2MD)kJ	3	CRV	20829
3	25	0	Inf. Dilution	dH:-961.350(5.1MD)kJ	3	CRV	20829

Reaction No. 74553

F2(g) + Zr(s) - 2<e-1> = Zr+4_F-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:208.9(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1197.4(9.5MD)kJ	0	CRV	20829
3	25	0	Inf. Dilution	dH:-1296.0(5.2MD)kJ	2	CRV	20829

Reaction No. 74554

1.5<F2(g)> + Zr(s) = ZrF3(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1065.7(1.7MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-1075.9(1.2MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:68.400(2.0MD)J/K	6	CRV	20829

Reaction No. 74555

1.5<F2(g)> + Zr(s) - e-1 = Zr+4_F-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:258.6(4SF)	0	ELC	
2	25	0	Inf. Dilution	dG:-1514.2(9.7MD)kJ	0	CRV	20829
3	25	0	Inf. Dilution	dH:-1625.8(5.6MD)kJ	2	CRV	20829

Reaction No. 74556

2<F2(g)> + Zr(s) = Zr+4_F-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:318.8(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1826.5(9.9MD)kJ	0	CRV	20829
3	25	0	Inf. Dilution	dH:-1971.9(6.2MD)kJ	2	CRV	20829

Reaction No. 74557

2<F2(g)> + Zr(s) = ZrF4(l)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1975.5(1.1MD)kJ	6	CRV	20829

Reaction No. 74558

2<F2(g)> + Zr(s) = ZrF4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1634.6(1.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-1671.1(1.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:87.440(0.5MD)J/K	6	CRV	20829

Reaction No. 74559

2.5<F2(g)> + Zr(s) + e-1 = Zr+4_F-1(5)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:372.5(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2133.5(10.0MD)kJ	0	CRV	20829

Reaction No. 74560

3<F2(g)> + Zr(s) + 2<e-1> = Zr+4_F-1(6)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:425.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2435.1(10.3MD)kJ	0	CRV	20829

Reaction No. 74561

0.5<Cl2(g)> + Zr(s) = ZrCl(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-264.480(2.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-291.200(2.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:47.160(0.6MD)J/K	6	CRV	20829

Reaction No. 74562

0.5<Cl2(g)> + Zr(s) = ZrCl(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:200.420(23.6MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:231.840(23.6MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:35.700(1.5MD)J/K	6	CRV	20829

Reaction No. 74563

0.5<Cl2(g)> + Zr(s) - 3<e-1> = Zr+4_Cl-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:116.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-668.800(9.2MD)kJ	0	CRV	20829

Reaction No. 74564

Cl2(g) + Zr(s) = ZrCl2(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-154.310(19.9MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-145.390(19.8MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:58.300(2.5MD)J/K	6	CRV	20829

Reaction No. 74565

Cl2(g) + Zr(s) - 2<e-1> = Zr+4_Cl-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:138.5(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-803.330(9.3MD)kJ	0	CRV	20829

Reaction No. 74566

1.5<Cl2(g)> + Zr(s) = ZrCl3(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-510.570(9.2MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-520.140(9.1MD)kJ	6	CRV	20829

3	25	0	Inf. Dilution	Cp:76.100(2.5MD)J/K	6	CRV	20829

Reaction No. 74567

2<Cl2(g)> + Zr(s) = ZrCl4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-836.430(0.9MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-871.500(0.9MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:98.200(1.6MD)J/K	6	CRV	20829

Reaction No. 74568

0.5<Br2(l)> + Zr(s) = ZrBr(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-180.(3SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-218.90(3.5MD)kJ	6	CRV	20829

Reaction No. 74569

0.5<Br2(l)> + Zr(s) = ZrBr(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	Cp:36.700(1.5MD)J/K	6	CRV	20829

Reaction No. 74570

Br2(l) + Zr(s) = ZrBr2(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	Cp:60.400(2.5MD)J/K	6	CRV	20829

Reaction No. 74571

1.5<Br2(l)> + Zr(s) = ZrBr3(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	Cp:79.400(2.5MD)J/K	6	CRV	20829

Reaction No. 74572

2<Br2(l)> + Zr(s) = ZrBr4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-664.670(3.8MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-643.500(3.7MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:102.700(1.0MD)J/K	6	CRV	20829

Reaction No. 74573

0.5<I2(s)> + Zr(s) = ZrI(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-150.420(1.2MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-152.400(1.2MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:50.800(0.1MD)J/K	6	CRV	20829

Reaction No. 74574

0.5<I2(s)> + Zr(s) = ZrI(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:349.320(11.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:403.000(11.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:37.200(1.5MD)J/K	6	CRV	20829

Reaction No. 74575

I2(s) + Zr(s) = ZrI2(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:77.761(16.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:135.000(16.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:61.400(2.5MD)J/K	6	CRV	20829

Reaction No. 74576

1.5<I2(s)> + Zr(s) = ZrI3(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-185.880(4.4MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-130.000(4.1MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:80.900(2.5MD)J/K	6	CRV	20829

Reaction No. 74577

2<I2(s)> + Zr(s) = ZrI4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-405.960(3.7MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-355.500(3.7MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:104.590(0.2MD)J/K	6	CRV	20829

Reaction No. 74578

3<S(s)> + Zr(s) = ZrS3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	dG:-601.920(4.8MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-615.200(4.6MD)kJ	6	CRV	20829

Reaction No. 74579

3<O2(g)> + 2<S(s)> + Zr(s) = Zr(SO3)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1834.9(10.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-2011.0(10.0MD)kJ	6	CRV	20829

Reaction No. 74580

2<O2(g)> + S(s) + Zr(s) - 2<e-1> = Zr+4_SO4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:229.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1312.7(9.2MD)kJ	0	CRV	20829
3	25	0	Inf. Dilution	dH:-1480.9(5.1MD)kJ	3	CRV	20829

Reaction No. 74581

4<O2(g)> + 2<S(s)> + Zr(s) = Zr+4_SO4-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:364.5(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2082.4(9.3MD)kJ	0	CRV	20829
3	25	0	Inf. Dilution	dH:-2359.8(5.2MD)kJ	2	CRV	20829

Reaction No. 74582

6<O2(g)> + 3<S(s)> + Zr(s) + 2<e-1> = Zr+4_SO4-2(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:496.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2842.1(9.7MD)kJ	0	CRV	20829

Reaction No. 74583

4<H2(g)> + 6<O2(g)> + 2<S(s)> + Zr(s) = Zr+4_SO4-2(2)_H2O(4)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3008.8(2.6MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3470.9(2.6MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:321.300(0.3MD)J/K	6	CRV	20829

Reaction No. 74584

3<O2(g)> + 2<Se(s)> + Zr(s) = Zr(SeO3)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1300.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-1609.6(2.7MD)kJ	6	CRV	20829

Reaction No. 74585

1.843<Te(s)> + Zr(s) = ZrTe1.843(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-291.330(7.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-295.000(7.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:70.590(0.3MD)J/K	6	CRV	20829

Reaction No. 74586

2<Te(s)> + Zr(s) = ZrTe2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-290.040(7.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-294.000(7.0MD)kJ	6	CRV	20829

Reaction No. 74587

4<Te(s)> + 5<Zr(s)> = Zr5Te4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-810.430(44.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-829.000(44.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:222.000(0.3MD)J/K	6	CRV	20829

Reaction No. 74588

4<O2(g)> + 3<Te(s)> + Zr(s) = ZrTe3O8(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1700.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-2119.2(9.6MD)kJ	6	CRV	20829

Reaction No. 74589

0.5<N2(g)> + Zr(s) = ZrN(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-341.850(1.9MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-370.500(1.4MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:39.700(1.4MD)J/K	6	CRV	20829

Reaction No. 74590

0.5<N2(g)> + 1.5<O2(g)> + Zr(s) - 3<e-1> = Zr+4_NO3-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:112.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-648.380(9.2MD)kJ	0	CRV	20829

Reaction No. 74591

N2(g) + 3<O2(g)> + Zr(s) - 2<e-1> = Zr+4_NO3-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-765.170(9.3MD)kJ	6	CRV	20829

Reaction No. 74592

3.5<O2(g)> + 2<P(s)> + Zr(s) = ZrP2O7(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	Cp:150.700(1.5MD)J/K	6	CRV	20829

Reaction No. 74593

2<H2(g)> + 4.5<O2(g)> + 2<P(s)> + Zr(s) = Zr(HPO4)2.H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3194.5(20.2MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3466.1(1.6MD)kJ	6	CRV	20829

Reaction No. 74594

3<H2(g)> + 5<O2(g)> + 2<P(s)> + Zr(s) = Zr(HPO4)2.2H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3000.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-3742.1(13.3MD)kJ	6	CRV	20829

Reaction No. 74595

2<As(s)> + 2<H2(g)> + 4.5<O2(g)> + Zr(s) = Zr(HAsO4)2.H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2100.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-2659.3(12.0MD)kJ	6	CRV	20829

Reaction No. 74596

4<C(s)> + 6<O2(g)> + Zr(s) + 4<e-1> = Zr+4_CO3-2(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	dG:-2885.0(10.9MD)kJ	6	CRV	20829

Reaction No. 74597

2<Ca(s)> + 6<O2(g)> + 3<Si(s)> + Zr(s) = Ca2ZrSi3O12(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5951.5(15.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-6283.0(15.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:252.500(2.5MD)J/K	6	CRV	20829

Reaction No. 74598

3<Ca(s)> + 4.5<O2(g)> + 2<Si(s)> + Zr(s) = Ca3ZrSi2O9(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-4769.3(10.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-5029.0(10.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:238.100(1.5MD)J/K	6	CRV	20829

Reaction No. 74599

3.5<O2(g)> + 2<Si(s)> + Sr(s) + Zr(s) = SrZrSi2O7(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3444.0(4.2MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3640.8(4.2MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:187.810(0.4MD)J/K	6	CRV	20829

Reaction No. 74600

9<O2(g)> + 5<Si(s)> + 6<Sr(s)> + Zr(s) = Sr6ZrSi5O18(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-7800.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-9867.9(8.6MD)kJ	6	CRV	20829

Reaction No. 74601

2<Na(s)> + 2.5<O2(g)> + Si(s) + Zr(s) = Na2ZrSiO5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2523.5(20.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-2670.0(20.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:172.400(2.0MD)J/K	6	CRV	20829

Reaction No. 74602

2<Na(s)> + 3.5<O2(g)> + 2<Si(s)> + Zr(s) = Na2ZrSi2O7(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3412.4(10.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3606.0(10.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:211.000(5.0MD)J/K	6	CRV	20829

Reaction No. 74603

2<H2(g)> + 2<Na(s)> + 5.5<O2(g)> + 3<Si(s)> + Zr(s) = Na2ZrSi3O9.2H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-4654.0(20.0MD)kJ	6	CRV	20829

Reaction No. 74604

4<O2(g)> + 2<S(s)> + Zr(s) = Zr+4_SO4-2(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2009.5(2.1MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-2245.2(2.1MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:185.100(10.0MD)J/K	6	CRV	20829

Reaction No. 74605

2<Na(s)> + 5.5<O2(g)> + 4<Si(s)> + Zr(s) = Na2ZrSi4O11(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5181.0(20.000MD)kJ	6	CRV	20829

Reaction No. 74606

3<H2(g)> + 2<Na(s)> + 9<O2(g)> + 6<Si(s)> + Zr(s) = Na2ZrSi6O15.3H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-7931.0(30.000MD)kJ	6	CRV	20829

Reaction No. 74607

4<Na(s)> + 6<O2(g)> + 3<Si(s)> + 2<Zr(s)> = Na4Zr2Si3O12(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5944.0(20.057MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-6258.0(20.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:365.000(5.0MD)J/K	6	CRV	20829

Reaction No. 74608

Na(s) + 6<O2(g)> + 3<P(s)> + 2<Zr(s)> = NaZr2P3O12(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-4557.0(20.2MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-4889.0(20.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:310.000(10.0MD)J/K	6	CRV	20829

Reaction No. 74609

2<Cs(s)> + 3.5<O2(g)> + 2<Si(s)> + Zr(s) = Cs2ZrSi2O7(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-3587.0(10.0MD)kJ	6	CRV	20829

Reaction No. 74610

1.5<S(s)> + Zr(s) = ZrS1.5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-502.110(3.9MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-509.400(1.1MD)kJ	6	CRV	20829

Reaction No. 74611

H2(g) + 4<O2(g)> + 2<P(s)> + Zr(s) = Zr(HPO4)2(a.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2987.0(23.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3166.2(6.0MD)kJ	6	CRV	20829

Reaction No. 74612

H2(g) + 4<O2(g)> + 2<P(s)> + Zr(s) = Zr(HPO4)2(b.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2500.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-3159.1(6.0MD)kJ	6	CRV	20829

Reaction No. 74613

2<As(s)> + H2(g) + 4<O2(g)> + Zr(s) = Zr(HAsO4)2(a.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1804.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-2364.7(12.1MD)kJ	6	CRV	20829

Reaction No. 74614

2<As(s)> + H2(g) + 4<O2(g)> + Zr(s) = Zr(HAsO4)2(b.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	dG:-1800.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-2360.9(12.0MD)kJ	6	CRV	20829

Reaction No. 74615

0.5<Zr(s)> + 0.5<ZrO2(s)> = ZrO(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:550.(3SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:606.500(27.0MD)kJ	6	CRV	20829

Reaction No. 74616

ZrO2(s) = ZrO2(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-130.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dH:811.000(47.0MD)kJ	4	CRV	20829

Reaction No. 74617

Zr+4 + 4<H2O> = Zr+4_OH-1(4)_(s) + 4<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.240(0.100MD)	6	CRV	20829
2	25	0	Inf. Dilution	lgK:2.77(0.13SD)	4	CRV	32222

Reaction No. 74618

Zr+4 + 6<H2O> = Zr+4_OH-1(6) + 6<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-33.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-29.000(0.700MD)	0	CRV	20829
3	25	0	Inf. Dilution	lgK:-31.96(0.35SD)	0	CRV	32222

Reaction No. 74619

3<Zr+4> + 9<H2O> = Zr+4(3)_OH-1(9) + 9<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.190(0.080MD)	3	CRV	20829

Note (1): Found not to be required by [18RKA_31367]

Reaction No. 74620

4<Zr+4> + 15<H2O> = Zr+4(4)_OH-1(15) + 15<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:12.580(0.240MD)	3	CRV	20829
Note (1): Found not to be required by [18RKA_31367]							
2	25	0	Inf. Dilution	lgK:0.04(0.52SD)	0	CRV	32222

Reaction No. 74621

4<Zr+4> + 16<H2O> = Zr+4(4)_OH-1(16) + 16<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.390(0.800MD)	3	CRV	20829
Note (1): Found not to be required by [18RKA_31367]							
2	25	0	Inf. Dilution	lgK:-4.36(0.61SD)	0	CRV	32222

Reaction No. 74622

Zr+4_F-1(4)_(s) = ZrF4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:240.180(0.040MD)kJ	3	CRV	20829

Reaction No. 74623

Zr+4_F-1(4)_(s) = ZrF4(l)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-64.220(0.420MD)kJ	3	CRV	20829

Reaction No. 74624

2<ZrCl3(g)> = ZrCl2(g) + ZrCl4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:23.4000(7.900MD)kJ	6	CRV	20829

Reaction No. 74625

Zr+4_Cl-1(4)_(s) = ZrCl4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.3(2SF)	1	ELC	
2	25	0	Inf. Dilution	dH:109.300(0.040MD)kJ	6	CRV	20829

Reaction No. 74626

Zr+4_Br-1(4)_(s) = ZrBr4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:116.000(1.800MD)kJ	6	CRV	20829

Reaction No. 74627

Zr+4_I-1(4)_(s) = ZrI4(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.122(0.064MD)	6	CRV	20829
2	25	0	Inf. Dilution	dH:133.400(0.300MD)kJ	6	CRV	20829

Reaction No. 74628							
0.5<N2(g)> + ZrO2(s) = O2(g) + ZrN(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:730.100(1.200MD)kJ	6	CRV	20829

Reaction No. 74629							
H2O + 2<H+1(3)_PO4-3> + Zr+4 = 4<H+1> + Zr(HPO4)2.H2O(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.800(3.100MD)	6	CRV	20829

Reaction No. 74630							
Zr+4 + 4<CO3-2> = Zr+4_CO3-2(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.900(1.000MD)	6	CRV	20829
2	25	0	Inf. Dilution	lgK:42.4(0.1SD)	4	MSL	29848
3	25	0	Inf. Dilution	lgK:42.(2SF)	2	EOD	29848

Reaction No. 74631							
Zr+4 + H+1(4)_CDTA-4 = Zr+4_H+1(4)_CDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:29.88(4SF)	5	MIX	13428
2	25	0	Inf. Dilution	lgK:29.7(3SF)	1	ESC	

Reaction No. 74632							
Zr+4 + H+1_Glycolic-1 = Zr+4_H+1_Glycolic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:7.7(2SF)	3	MKN	7753

Reaction No. 74633							
Zr+4 + H+1_PhenAcet-1 = Zr+4_H+1_PhenAcet-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:6.6(2SF)	3	MKN	7753

2	25	0.01	Unknown	lgK:6.8(2SF)	3	MKN	7753
3	25	0.1	Unknown	lgK:6.2(2SF)	3	MKN	7753

Reaction No. 74634

Zr+4 + SCN-1 = Zr+4_SCN-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-2.(1SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-3.8(2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-2.6(3SF)	2	EAE	

Reaction No. 74635

Zr+4_SCN-1 + SCN-1 = Zr+4_SCN-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-1.4(2SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-3.5(2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-2.6(3SF)	2	EAE	

Reaction No. 74636

Zr+4_SCN-1(2) + SCN-1 = Zr+4_SCN-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-1.25(3SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-3.4(2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-2.8(3SF)	2	EAE	

Reaction No. 74637

Zr+4_SCN-1(3) + SCN-1 = Zr+4_SCN-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-1.1(2SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-3.2(2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-2.9(3SF)	2	EAE	

Reaction No. 74638

Zr+4_SCN-1(4) + SCN-1 = Zr+4_SCN-1(5)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-1.1(2SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-3.1(2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-3.1(3SF)	2	EAE	

Reaction No. 74639							
Zr+4_SCN-1(5) + SCN-1 = Zr+4_SCN-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-1.0(2SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-2.9(2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-3.2(3SF)	2	EAE	

Reaction No. 74640							
Zr+4_SCN-1(6) + SCN-1 = Zr+4_SCN-1(7)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-0.98(2SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-2.8(2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-3.4(3SF)	2	EAE	

Reaction No. 74641							
Zr+4_SCN-1(7) + SCN-1 = Zr+4_SCN-1(8)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	HClO4	lgK:-0.96(2SF)	0	MSP	13450
2	23	0.8	HClO4	lgK:-2.7(2SF)	4	MSP	13450
3	25	0	Inf. Dilution	lgK:-3.61(3SF)	2	EAE	

Reaction No. 74642							
Zr+4 + Malic-2 = Zr+4_Malic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:1.9(2SF)	3	MCE	7744
2	25	0	Inf. Dilution	lgK:3.84(3SF)	2	EAE	

Reaction No. 74643							
Zr+4 + Lactic-1 = Zr+4_Lactic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:2.27(3SF)	5	MCE	3262
2	23	2	HClO4	lgK:2.0(2SF)	3	MCE	7744
3	25	0	Inf. Dilution	lgK:3.17(3SF)	2	EAE	

Reaction No. 74644
Zr+4 + Tartaric-2 = Zr+4_Tartaric-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HCl	lgK:6.86(3SF)	3	MLC	4433
2	23	2	HClO4	lgK:2.2(2SF)	3	MCE	7744

Reaction No. 74645

Citric-3 + Zr+4 = Zr+4_Citric-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HCl	lgK:10.78(4SF)	3	MLC	4433
2	23	2	HClO4	lgK:3.1(2SF)	0	MCE	7744
3	25	0	Inf. Dilution	lgK:13.27(4SF)	4	CRV	31301

Reaction No. 74646

Zr+4 + 234TriOHPentDioic-2 = Zr+4_234TriOHPentDioic-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:3.41(3SF)	5	MCE	3262
2	23	2	HClO4	lgK:3.1(2SF)	3	MCE	7744

Reaction No. 74647

Zr+4 + H+1(6)_EDTA-4 = Zr+4_EDTA-4 + 6<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	HClO4	dH:-5.86(0.15MD)	2	MCL	5106
2	25	3	HClO4	dH:-5.72(0.18MD)	2	MCL	5106
3	25	4	HClO4	dH:-5.66(0.16MD)	2	MCL	5106

Reaction No. 74648

Zr+4_Lactic-1 + Lactic-1 = Zr+4_Lactic-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:2.54(3SF)	5	MCE	3262
2	25	0	Inf. Dilution	lgK:3.27(3SF)	2	EAE	

Reaction No. 74649

Zr+4 + H+1_Malic-2 = Zr+4_H+1_Malic-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:2.23(3SF)	5	MCE	3262
2	25	0	Inf. Dilution	lgK:3.2(3SF)	2	EAE	

Reaction No. 74650

Zr+4_234TriOHPentDioic-2 + 234TriOHPentDioic-2 = Zr+4_234TriOHPentDioic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgK:5.40(3SF)	5	MCE	3262

Reaction No. 74651							
Zr+4 + Glu-2 = Zr+4_Glu-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:9.60(3SF)	4	MGL	22615

Reaction No. 74652							
Zr+4_Glu-2 + Glu-2 = Zr+4_Glu-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.40(3SF)	4	MGL	22615

Reaction No. 74653							
Zr+4_Glu-2(2) + Glu-2 = Zr+4_Glu-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:3.32(3SF)	4	MGL	22615

Reaction No. 74654							
Zr+4 + H+1(4)_EDTA-4 = Zr+4_EDTA-4 + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.3(3SF)	1	ELC	
2	25	0.01	HClO4	lgR:19.4(3SF)	3	MMX	13427

Reaction No. 74855							
Zr+4 + e-1 = Zr+3 Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-96.0(3SF)	0	EOD	22560
Note (1): From Latimer WM, The Oxidation States of the Elements, 1952							
2	25	0	Inf. Dilution	E:-1.430(4SF)	0	MEF	22560
Note (2): From Latimer WM, The Oxidation States of the Elements, 1952							

Reaction No. 75034							
CrO4-2 + Zr+4 = Zr+4_CrO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	0.01	0	Inf. Dilution	lgK:7.0102(5SF)	3	EOD	22958
2	25	0	Inf. Dilution	lgK:7.3125(5SF)	3	EOD	22958
3	60	0	Inf. Dilution	lgK:7.9602(5SF)	3	EOD	22958
4	100	0	Inf. Dilution	lgK:8.8346(5SF)	3	EOD	22958
5	150	0	Inf. Dilution	lgK:10.048(5SF)	3	EOD	22958
6	200	0	Inf. Dilution	lgK:11.3930(6SF)	3	EOD	22958

Reaction No. 75792

Zr+3 + Asn-1 = Zr+3_Asn-1

Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:8.80(3SF)	0	MGL	22630

Reaction No. 75793

Zr+3_Asn-1 + Asn-1 = Zr+3_Asn-1(2)

Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.25(3SF)	0	MGL	22630

Reaction No. 75794

Zr+3 + 2<Asn-1> = Zr+3_Asn-1(2)

Note: Zirconium exists in aqueous solutions in the + 4 oxidation state only. See Brown et al., OECD Chem. Thermodynamics Ser., Vol. 8, 2005, p. 97

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:15.05(4SF)	0	MGL	22630

Reaction No. 75821

Zr+4 + 2<Glu-2> = Zr+4_Glu-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:16.00(4SF)	4	MGL	22615

Reaction No. 75847

Zr+4 + 2<Asp-2> = Zr+4_Asp-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:16.55(4SF)	4	MGL	22615

Reaction No. 75856

Zr+4 + 2<Gln-1> = Zr+4_Gln-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:14.85(4SF)	4	MGL	22630

Reaction No. 76500

Zr+4 + 6<OH-1> = Zr+4_OH-1(6)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55.(2SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:48.(2SF)	0	EOD	29848
3	25	0	Inf. Dilution	lgK:50.72(0.24SD)	4	CRV	31367
Note (3): SIT fit							
4	25	0	Inf. Dilution	lgK:50.95(0.35SD)	4	CRV	31367
Note (4): Pitzer fit							

Reaction No. 78022

2<EDTA-4> + 2<Zr+4> = Zr+4(2)_EDTA-4(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:70.0(4SF)	1	ELC	

Reaction No. 78032

Citric-3 + Zr+4 + H+1 = Zr+4_H+1_Citric-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.9(3SF)	1	ELC	

Reaction No. 78042

Citric-3 + Zr+4 + 2<H+1> = Zr+4_H+1(2)_Citric-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.27(4SF)	1	ELC	

Reaction No. 78045

Gly-1 + Zr+4 + H+1 = Zr+4_H+1_Gly-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.8(4SF)	1	ELC	

Reaction No. 78114

EDTA-4 + Zr+4 + OH-1 = Zr+4_OH-1_EDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:40.84(4SF)	1	ELC	

Reaction No. 78164

2<EDTA-4> + 2<Zr+4> + 2<OH-1> = Zr+4(2)_OH-1(2)_EDTA-4(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:84.97(4SF)	1	ELC	

Reaction No. 78196

Acetic-1 + Zr+4 + 2<OH-1> = Zr+4_Acetic-1_OH-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.13(4SF)	1	ELC	

Reaction No. 78198

Formic-1 + Zr+4 + 3<OH-1> = Zr+4_Formic-1_OH-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:44.73(4SF)	1	ELC	

Reaction No. 78212

Acetic-1 + Zr+4 + 3<OH-1> = Zr+4_Acetic-1_OH-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:44.68(4SF)	1	ELC	

Reaction No. 78225

Propanoic-1 + Zr+4 + 3<OH-1> = Zr+4_OH-1(3)_Propanoic-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:45.13(4SF)	1	ELC	

Reaction No. 78226

2<Acetic-1> + Zr+4 + 3<OH-1> = Zr+4_Acetic-1(2)_OH-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:46.51(4SF)	1	ELC	

Reaction No. 78277

Oxalic-2 + Zr+4 + H+1 = Zr+4_H+1_Oxalic-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.84(4SF)	1	ELC	

Reaction No. 78328

2<Oxalic-2> + Zr+4 + 2<H+1> = Zr+4_H+1(2)_Oxalic-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.76(4SF)	1	ELC	

Reaction No. 78412

4<Oxalic-2> + Zr+4 + 4<H+1> = Zr+4_H+1(4)_Oxalic-2(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.31(4SF)	1	ELC	

Reaction No. 78626

Ala-1 + Zr+4 + H+1 = Zr+4_H+1_Ala-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.04(4SF)	1	ELC	

Reaction No. 78632

EDTA-4 + Zr+4 + H+1 = Zr+4_H+1_EDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:39.35(4SF)	1	ELC	

Reaction No. 80690

ZrO2(am.,s) + 2<H2O> = Zr+4 + 4<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-56.19(0.03SD)	3	CRV	31367

Reaction No. 80691

Zr+4 + 6<OH-1> + 3<Ca+2> = Zr+4_Ca+2(3)_OH-1(6)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:57.00(0.22SD)	4	CRV	31367
Note (1): SIT fit							
2	25	0	Inf. Dilution	lgK:57.47(0.20SD)	4	CRV	31367
Note (2): Pitzer fit							

Reaction No. 81029

Zr+4 + 4<H2O> = Zr+4_OH-1(4)_(am.,s) + 4<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.14(0.13SD)	4	CRV	32222

Reaction No. 81334

BaZrO3(s) + 6<H+1> = Ba+2 + Zr+4 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.45(4SF)	4	CRV	32224

Reaction No. 81343

CaO.ZrO2(s) + 6<H+1> = Ca+2 + Zr+4 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.54(4SF)	4	CRV	32224

Reaction No. 81614

SrZrO3(s) + 6<H+1> = Sr+2 + Zr+4 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:29.28(4SF)	0	CRV	32224

Reaction No. 81707

ZrH2(s) = Zr+4 + 2<H+1> + 6<e-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:70.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:78.88(4SF)	0	CRV	32224

Reaction No. 81708

3<Zr+4> + 5<H2O> = Zr+4(3)_OH-1(5) + 5<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.7(2SF)	2	CRV	32224

Reaction No. 81709

ZrSiO4(s) + 4<H+1> = Zr+4 + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-14.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-5.64(3SF)	0	CRV	32224

Data Assessment

Weight	Assessment

9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CMN	Several experimental values (crit.)
CRV	Critical review
CSM	Smith & Martell Vols 1-6 (crit.)
EAE	Automated extrapolation
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
EIU	Critical IUPAC review (estim.)
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
ERS	Rough subjective
ESC	Standard conditions anchor
MCE	Cation exchange
MCG	Chromatogaphy, Gel
MCL	Calorimetry
MDG	Combination of Thermodynamic Data
MDS	Distribution between two phases
MEF	Electromotoric force, not specified

MGL	Glass electrode
MHY	Emf with hydrogen electrode
MIS	Ion selective electrode
MIX	Ion exchange
MKN	Rate of reaction
MLC	Ligand competition
MMX	Mixed techniques
MPL	Polarography
MPT	Potentiometry
MQH	Quinhydrone electrode
MRX	Emf with redox electrode
MSC	Miscellaneous
MSE	Solvent Extraction
MSL	Solubility
MSP	Spectroscopy
MTD	Temperature difference

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